PUFormer: A Physics-Unfolded Transformer With Exact Observation Operators for Hyperspectral and Multispectral Image Fusion

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***Abstract*—Deep unfolding networks for hyperspectral and multispectral image fusion alternate data-consistency steps with learned priors, but their observation operators are learned or approximated, and they are trained for one sensor pair. We formulate fusion as maximum a posteriori estimation with parametric blur and spectral-response operators, unroll its half-quadratic splitting, and establish three properties that make the operators usable inside the network. First, the two operators commute, and on any finite set of candidate parameters the true pair is identified by the two observations alone for almost every scene. Second, with a separable blur and the true border rule, the operators factor along the three axes of the image cube: the data step then has a closed form in a product eigenbasis, and the joint null space, the part of the image that no observation constrains, is explicit; it holds 88% of the unknowns for Chikusei. Third, an exact data step leaves the ground truth fixed, and under an inexact operator it moves it by an amount bounded independently of the learned weights. PUFormer builds on these results: a closed-form data step whose per-component gains are learned without losing these guarantees, a residual transformer prior, and training over a family of blurs, responses and noise levels with the operators estimated at test time. Against five published networks retrained from their authors' code under one budget, PUFormer is the most accurate on Chikusei and Pavia Centre, by 0.99 and 1.00 dB PSNR; its error in the observed subspace is 5.8–8.6 times smaller, and the null-space split attributes its margin to consistency and to the prior. Trained over operator families, it keeps its accuracy under changed blurs, responses and noise; outside the family and under misregistration it fails, as the other methods do. Code, logs and all results are public.**

***Index Terms*—Deep unfolding, hyperspectral image fusion, identifiability, Kronecker structure, null space, observation model, super-resolution.**

I. Introduction

HYPERSPECTRAL imagers record hundreds of narrow spectral bands but, for a fixed photon budget, at a coarse spatial resolution; multispectral imagers do the opposite. Hyperspectral and multispectral image fusion recovers a high-resolution hyperspectral image (HR-HSI) from one observation of each kind [1]–[3]. Classical methods formulate fusion as a regularized inverse problem, using component substitution [4], coupled matrix factorization [5], subspace regularization [6] or fast Sylvester-equation solvers [7]. Deep networks have since become the dominant approach, from convolutional reconstruction networks [8], [9] to transformers [10]–[13] and diffusion models [14]–[17].

Model-guided and unfolded networks [18]–[23] bring the observation model back into the network: each stage takes a step that reduces the mismatch between the current estimate and the two observations, and a learned module then refines the estimate. How faithfully the network reproduces the observation model has received less attention than the design of the learned module. In most of these networks the blur–decimation and spectral operators are learned convolution layers, or are implemented with the zero padding that convolution layers apply at the image border. The benchmark data, however, are simulated with a fixed blur and a different border rule. The network's operator is then not the one that produced the data, and even the ground truth fails the network's own consistency check. In simulated benchmarks this mismatch is avoidable, because the forward model is known exactly. A second difficulty concerns evaluation: reported numbers on the same dataset vary widely across papers because of differences in sensor response, blur, data split, normalization and even metric definitions [3], so improvements are only meaningful within one fully specified protocol in which all methods are run.

This paper treats the observation model inside an unfolded network as an object to be analyzed, estimated and exploited rather than approximated. We write fusion as maximum a posteriori estimation with parametric operators, unroll its half-quadratic splitting, and derive the properties that the operators of such a network have; the network, PUFormer, is built from them. Our contributions are:

1) **Theory of the observation operators in unfolded fusion.** We prove that the blur–decimation and spectral operators commute and that their parameters are identified by the two observations alone for almost every scene (Theorem 1); that with a separable blur and the true border rule they factor along the three axes of the image cube, which gives the data step in closed form and the joint null space explicitly (Theorem 2); and that an exact data step leaves the ground truth fixed, while an inexact operator moves it by an amount that the exact proximal step bounds independently of the learned weights, with a corollary for border rules (Theorem 3).
2) **An unfolded network built on these results.** PUFormer uses the closed-form exact data step, a learned spectral data step that reweights its eigencomponents while provably keeping the fixed point and nonexpansiveness

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Code, training logs and all result files are available at https://github.com/Amarnath10i/Research-On-Hyperspectral-ImageFussion. Supplementary material (training stages, robustness, metric analyses, per-image results and complete benchmark metrics) accompanies this paper.

(Proposition 1), observation-model conditioning of the step sizes and the prior, and training over a family of blurs, spectral responses and noise levels with the operators estimated at test time.

3) **An evaluation that explains the results.** Five published networks are retrained from their authors' code under one budget and also with their own recipes; the null-space split of the error shows how much of each method's error contradicts the observations and attributes PUFormer's margin; ablations, tests inside and outside the operator family, and further datasets and scale factors delimit what the exact operators do and do not achieve.

II. Related Work

A. Model-based fusion

Component substitution methods such as GSA [4] inject multispectral detail into the upsampled hyperspectral bands. Unmixing-based methods such as CNMF [5] factor both observations with shared endmembers, and subspace methods such as HySure [6] regularize the HR-HSI within a low-dimensional spectral subspace, whose dimension can be estimated with HySime [24]. FUSE [7] solves the resulting quadratic problem in closed form through a Sylvester equation. These methods use the forward model explicitly but rely on hand-crafted priors.

B. Supervised deep fusion

SSR-NET [8] and GuidedNet [9] learn a direct mapping from the two observations. Transformers model long-range spatial and spectral dependencies: PSRT [10] uses pyramid shuffle-and-reshuffle attention, MIMO-SST [11] multi-input multi-output supervision, LGCT [12] local–global collaboration, DSPNet [25] a dual spatial–spectral pyramid, SSDT [13] dilated attention, and MIMFormer [26], AMSF-Net [27], ASMNet [28] and DPFormer [29] multiscale spectral–spatial designs. Recent work also studies state-space models [30], [31], mixture-of-experts routing [32] and modality interaction [33]. General image-fusion frameworks such as U2Net [34] and OTPNet [35] have also been applied to this task.

C. Model-guided and unfolded networks

Algorithm unrolling [36]–[38] turns the iterations of an optimization method into network stages. MHF-Net [18], MoG-DCN [19] and DHIF-Net [20] unfold fusion-specific solvers, MHF-Net and DHIF-Net with operators learned as convolution layers, DUNET [21] adds transformer priors for unregistered inputs, HMPNet [22] extends unfolding to three modalities, and an information-aware transformer-based unfolding network (ITU-Net) [23], a mutual-guidance unfolding network (SMGU) [39], the perceptive spectral transformer unfolding network (PSTUN) [40], which also handles arbitrary scales, and the cross-domain-aware unfolding transformer CaFormer [41] strengthen the learned proximal steps with transformer priors. PUFormer belongs to this family, and none of its prior's components is new: passing features from one stage to the next follows memory-augmented unfolding [42], spectral-wise attention inside an unfolded network appears in DAUHST [43], and the transposed attention is that of Restormer [44]. A single unfolded network that takes the blur kernel as an input, and therefore handles kernels it was not trained on, was shown for single-image super-resolution by USRNet [45].

D. Boundary conditions

In classical deconvolution, an image border handled differently by the model and by the data causes ringing that spreads inward, and reflexive (Neumann) boundary conditions, i.e., reflect padding, are a standard remedy [46]–[48]. In iterative reconstruction, a back-projection that is not the exact adjoint of the forward model changes the point to which the iteration converges [49], [50], and in convolutional networks zero padding is itself a source of spatial bias near the border [51]. In unfolded fusion networks the operators are either learned or implemented with the zero padding of convolution layers, while benchmark data are simulated with other border rules; to our knowledge, the effect of this mismatch has not been examined. Our contribution is on this side: an unfolded network whose operators, border included, are exactly those of the simulation, and an ablation that separates the exact operator from a zero-border and a learned operator under the same budget.

E. Generative fusion

Diffusion models have been conditioned on the two observations (DDIF [14]), constrained by the observation model during sampling (CDPS [16]), restricted to a learned subspace (ARGS-Diff [17]) or combined with differential attention in a two-stage conditioning scheme [15]. Null-space formulations [52] make the data-consistency part of the reconstruction exact. The two-stage conditional diffusion model of Chen *et al.* [15] holds the best published Chikusei $\times 4$ results in our survey and is our main reference; its comparison also includes the correlation-aware network CLSNet [53] and the dual cross transformer (DCTransformer) [54], denoted DCT in their table.

III. Problem Formulation and Analysis

A. Observation model

Let $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$ be the HR-HSI with B bands ($B = 128$ for Chikusei and 92 for Pavia Centre). The two observations are

$$\mathbf{Y}_{\mathrm{H}} = \mathcal{D}_\theta \mathbf{X} + \mathbf{N}_{\mathrm{H}}, \qquad \mathbf{Y}_{\mathrm{M}} = \mathcal{R}_\phi \mathbf{X} + \mathbf{N}_{\mathrm{M}}, \tag{1}$$

where $\mathcal{D}_\theta = \mathcal{S}\,\mathcal{K}_\theta\,\mathcal{P}$ applies, band by band, reflect padding $\mathcal{P}$ of width p, a valid correlation $\mathcal{K}_\theta$ with a normalized $(2p+1)\times(2p+1)$ Gaussian kernel of standard deviation θ, and decimation $\mathcal{S}$ that keeps every s-th pixel starting at index zero, so $\mathbf{Y}_{\mathrm{H}} \in \mathbb{R}^{B\times H/s\times W/s}$; $\mathcal{R}_\phi$ applies to every pixel the M spectral responses of the multispectral sensor, shifted along wavelength by ϕ nm; and $\mathbf{N}_{\mathrm{H}}$, $\mathbf{N}_{\mathrm{M}}$ are independent white Gaussian noise of standard deviation ς_{H} and ς_{M}. The benchmark protocol is the special case $\theta = \sigma_0$, $\phi = 0$ without noise,

with $s = 4$, $p = 3$ and $\sigma_0 = 2$, the 7×7 blur of [15]; we write $\mathcal{D} = \mathcal{D}_{\sigma_0}$ and $\mathcal{R} = \mathcal{R}_0$ for the nominal operators and keep s and p general where the analysis does not depend on them. The spectral operator $\mathcal{R} \in \mathbb{R}^{M \times B}$ integrates the spectrum with the $M = 8$ WorldView-2 band responses [55] for Chikusei and with the $M = 4$ IKONOS band responses [56] for Pavia Centre. Because the manufacturers publish the responses only as plots together with band edges, we model each band as a flat top with logistic flanks (4 nm roll-off) that pass exactly through the published edges, at 5% response for WorldView-2, whose edges are tabulated at that level, and at 50% for the IKONOS band limits (blue 445–516, green 506–595, red 632–698 and near infrared 757–853 nm), and normalize every band to unit sum.

Exact adjoint. Every gradient step on $\|\mathcal{D}\mathbf{X} - \mathbf{Y}_\mathrm{H}\|^2$ requires $\mathcal{D}^\top = \mathcal{P}^\top \mathcal{K}^\top \mathcal{S}^\top$. $\mathcal{S}^\top$ inserts zeros and $\mathcal{K}^\top$ is a full convolution; together they are a strided transposed convolution with an output padding of $s-1$. The adjoint of reflect padding folds the padded margins back onto the pixels they were copied from. For a padded row index $j \in [0, H + 2p)$ the source row is $p - j$ for $j < p$, $j - p$ in the interior and $2H - 2 + p - j$ in the bottom margin, hence

$$(\mathcal{P}^\top \mathbf{g})_i = \mathbf{g}_{i+p} + [1 \le i \le p]\, \mathbf{g}_{p-i} + [H - 1 - p \le i \le H - 2]\, \mathbf{g}_{2H-2+p-i}, \tag{2}$$

applied to rows and then to columns. With (2), $\mathcal{D}$ in the network is exactly the simulation operator, and the inner-product test $\langle \mathcal{D}\mathbf{X}, \mathbf{Y} \rangle = \langle \mathbf{X}, \mathcal{D}^\top \mathbf{Y} \rangle$ holds to a relative error below 10^{-13} in double precision at every scale factor. $\mathcal{R}^\top$ is the transpose of the response matrix.

B. MAP estimation and its unfolding

For given operator parameters and noise levels, and a prior $p(\mathbf{X}) \propto \exp(-\lambda \Phi(\mathbf{X}))$, the maximum a posteriori estimate is

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \; \tfrac{1}{2\varsigma_\mathrm{H}^2} \|\mathcal{D}_\theta \mathbf{X} - \mathbf{Y}_\mathrm{H}\|^2 + \tfrac{1}{2\varsigma_\mathrm{M}^2} \|\mathcal{R}_\phi \mathbf{X} - \mathbf{Y}_\mathrm{M}\|^2 + \lambda \Phi(\mathbf{X}). \tag{3}$$

Half-quadratic splitting [45] introduces an auxiliary image $\mathbf{Z}$ coupled to $\mathbf{X}$ by a penalty $\frac{\mu_k}{2}\|\mathbf{Z} - \mathbf{X}\|^2$ and alternates, for $k = 1, \dots, K$, a data step and a prior step:

$$\mathbf{Z}_k = \arg\min_{\mathbf{Z}} \; \tfrac{1}{2\varsigma_\mathrm{H}^2} \|\mathcal{D}_\theta \mathbf{Z} - \mathbf{Y}_\mathrm{H}\|^2 + \tfrac{1}{2\varsigma_\mathrm{M}^2} \|\mathcal{R}_\phi \mathbf{Z} - \mathbf{Y}_\mathrm{M}\|^2 + \tfrac{\mu_k}{2} \|\mathbf{Z} - \mathbf{X}_{k-1}\|^2, \tag{4}$$

$$\mathbf{X}_k = \mathrm{prox}_{(\lambda/\mu_k)\Phi}(\mathbf{Z}_k). \tag{5}$$

With $\alpha_k = 1/(\mu_k \varsigma_\mathrm{H}^2)$ and $\beta_k = 1/(\mu_k \varsigma_\mathrm{M}^2)$, the data step (4) is the symmetric positive definite linear system

$$(\mathbf{I} + \alpha_k \mathcal{D}_\theta^\top \mathcal{D}_\theta + \beta_k \mathcal{R}_\phi^\top \mathcal{R}_\phi)\, \mathbf{Z}_k = \mathbf{X}_{k-1} + \alpha_k \mathcal{D}_\theta^\top \mathbf{Y}_\mathrm{H} + \beta_k \mathcal{R}_\phi^\top \mathbf{Y}_\mathrm{M}, \tag{6}$$

which conjugate gradients solve with the exact operators and adjoints alone, and a single gradient step from $\mathbf{X}_{k-1}$ gives its linearization

$$\mathbf{Z}_k \approx \mathbf{X}_{k-1} - \alpha_k \mathcal{D}_\theta^\top (\mathcal{D}_\theta \mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{H}) - \beta_k \mathcal{R}_\phi^\top (\mathcal{R}_\phi \mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{M}). \tag{7}$$

An unfolded network replaces the proximal map (5) by a learned module. Three properties of this derivation shape the network of Section IV. First, α_k and β_k are not independent design parameters: their ratio $\beta_k/\alpha_k = \varsigma_\mathrm{H}^2/\varsigma_\mathrm{M}^2$ weights the two observations by their noise. Second, the proximal map depends on λ/μ_k and hence on the noise level, so a learned prior should be told it. Third, the operators enter only through $\mathcal{D}_\theta$, $\mathcal{R}_\phi$ and their adjoints in (6), so operators known or estimated at test time can be used without retraining. Theorems 1 and 3 state when the operators can be identified from the observations and what an inexact operator costs.

C. Identifiability of the operators

Theorem 1 (Commutation and identifiability): Let $\mathcal{D}_\theta$ apply the same padding, blur and decimation to every band and $\mathcal{R}_\phi$ the same matrix $\mathbf{r}_\phi \in \mathbb{R}^{M \times B}$ to every pixel. (a) $\mathcal{D}_\theta \mathcal{R}_{\phi'} = \mathcal{R}_{\phi'} \mathcal{D}_\theta$ for all θ, ϕ', with $\mathcal{D}_\theta$ acting on M or B bands. Hence, for noise-free observations generated with (θ^*, ϕ^*), the criterion

$$J(\theta, \phi) = \|\mathcal{D}_\theta \mathbf{Y}_\mathrm{M} - \mathcal{R}_\phi \mathbf{Y}_\mathrm{H}\|^2 \tag{8}$$

vanishes at (θ^*, ϕ^*). (b) Let $\mathcal{G}$ be a finite set of candidate parameters on which $\theta \mapsto \mathcal{D}_\theta$ and $\phi \mapsto \mathbf{r}_\phi$ are injective, and let no column of $\mathbf{r}_\phi$ vanish. Then, for almost every $\mathbf{X}$ (with respect to the Lebesgue measure), (θ^*, ϕ^*) is the unique zero of J on $\mathcal{G}$.

Proof: (a) On vectorized images, $\mathcal{D}_\theta = \mathbf{I}_B \otimes \bar{\mathcal{D}}_\theta$ and $\mathcal{R}_\phi = \mathbf{r}_\phi \otimes \mathbf{I}_N$, with $\bar{\mathcal{D}}_\theta$ the single-band spatial operator and $N = HW$; by the mixed-product rule, $(\mathbf{I}_M \otimes \bar{\mathcal{D}}_\theta)(\mathbf{r}_\phi \otimes \mathbf{I}_N) = \mathbf{r}_\phi \otimes \bar{\mathcal{D}}_\theta = (\mathbf{r}_\phi \otimes \mathbf{I}_{N/s^2})(\mathbf{I}_B \otimes \bar{\mathcal{D}}_\theta)$. With $\mathbf{Y}_\mathrm{M} = \mathcal{R}_{\phi^*}\mathbf{X}$ and $\mathbf{Y}_\mathrm{H} = \mathcal{D}_{\theta^*}\mathbf{X}$, $\mathcal{D}_{\theta^*}\mathbf{Y}_\mathrm{M} = \mathcal{R}_{\phi^*}\mathcal{D}_{\theta^*}\mathbf{X} = \mathcal{R}_{\phi^*}\mathbf{Y}_\mathrm{H}$. (b) By (a), $J(\theta, \phi) = \|\mathbf{E}_{\theta\phi}\mathbf{X}\|^2$ with $\mathbf{E}_{\theta\phi} = \mathbf{r}_{\phi^*} \otimes \bar{\mathcal{D}}_\theta - \mathbf{r}_\phi \otimes \bar{\mathcal{D}}_{\theta^*}$. Suppose $\mathbf{E}_{\theta\phi} = \mathbf{0}$ and apply it to $\mathbf{X} = \mathbf{e}_b \otimes \mathbf{u}$, a single band b with spatial pattern $\mathbf{u}$: $(\mathbf{r}_{\phi^*})_{:b} \otimes \bar{\mathcal{D}}_\theta \mathbf{u} = (\mathbf{r}_\phi)_{:b} \otimes \bar{\mathcal{D}}_{\theta^*}\mathbf{u}$ for all $\mathbf{u}$. If $\theta \neq \theta^*$, then $\bar{\mathcal{D}}_\theta \neq c\,\bar{\mathcal{D}}_{\theta^*}$ for every scalar c: both kernels sum to one and reflect padding preserves constants, so both operators map a constant image to the same constant, which forces $c = 1$ and contradicts injectivity. Two linear maps of rank greater than one that are not scalar multiples of each other map some $\mathbf{u}$ to linearly independent images, and for this $\mathbf{u}$ the identity above forces $(\mathbf{r}_{\phi^*})_{:b} = (\mathbf{r}_\phi)_{:b} = \mathbf{0}$, a contradiction. If $\theta = \theta^*$, the identity gives $(\mathbf{r}_{\phi^*})_{:b} = (\mathbf{r}_\phi)_{:b}$ for every b, so $\phi = \phi^*$. Hence $\mathbf{E}_{\theta\phi} \neq \mathbf{0}$ for every $(\theta, \phi) \neq (\theta^*, \phi^*)$ in $\mathcal{G}$; its null space is a proper subspace and has measure zero, and so does the finite union of these null spaces. For every $\mathbf{X}$ outside it, $J(\theta, \phi) > 0$ whenever $(\theta, \phi) \neq (\theta^*, \phi^*)$. ∎

We therefore estimate (θ, ϕ) for each test pair by minimizing (8) on a grid (steps of $0.025\,\sigma_0$ and 1 nm), using only the two observations; HySure [6] uses the same relation to estimate the responses of a model-based method. With noise, $\mathbb{E}\, J(\theta, \phi) = \|\mathbf{E}_{\theta\phi}\mathbf{X}\|^2 + \frac{N}{s^2}\big(M\varsigma_\mathrm{M}^2\|\mathbf{k}_\theta\|^2 + \varsigma_\mathrm{H}^2\|\mathbf{r}_\phi\|_F^2\big)$ up to border terms, where $\mathbf{k}_\theta$ is the blur kernel. The noise term decreases with θ and barely depends on ϕ, so noise biases $\hat{\theta}$ upward by an amount that vanishes as the SNR grows. Theorem 1 also requires the family to contain the true operator; Section VI-H tests anisotropic and defocus blurs and misregistered observations, which it does not.

D. Separable structure: a closed-form data step and the fusion null space

A Gaussian kernel, isotropic or aligned with the image axes, is separable, and reflect and zero padding treat rows and columns independently; decimation does too. The blur–decimation operator therefore factors exactly, border included: for the cube $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$,

$$\mathcal{D}_\theta \mathbf{X} = \mathbf{X} \times_2 \mathbf{D}_\mathrm{r} \times_3 \mathbf{D}_\mathrm{c}, \qquad \mathcal{R}_\phi \mathbf{X} = \mathbf{X} \times_1 \mathbf{r}_\phi, \tag{9}$$

where $\times_n$ is the mode-n product and $\mathbf{D}_\mathrm{r} \in \mathbb{R}^{h\times H}$ and $\mathbf{D}_\mathrm{c} \in \mathbb{R}^{w\times W}$ pad, blur and decimate a single axis. Separable operators of this form are used by tensor-factorization fusion methods [57] to identify low-rank models, under periodic borders; here they yield an exact, closed-form data step and an explicit description of what fusion cannot observe.

Theorem 2 (Diagonalization and null space): Let $\mathbf{D}_\mathrm{r}^\top \mathbf{D}_\mathrm{r} = \mathbf{U}_\mathrm{r}\mathrm{diag}(\boldsymbol{\lambda}^\mathrm{r})\mathbf{U}_\mathrm{r}^\top$, $\mathbf{D}_\mathrm{c}^\top \mathbf{D}_\mathrm{c} = \mathbf{U}_\mathrm{c}\mathrm{diag}(\boldsymbol{\lambda}^\mathrm{c})\mathbf{U}_\mathrm{c}^\top$ and $\mathbf{r}_\phi^\top \mathbf{r}_\phi = \mathbf{V}\mathrm{diag}(\boldsymbol{\nu})\mathbf{V}^\top$ be eigendecompositions, and $\mathcal{Q}(\cdot) = (\cdot) \times_1 \mathbf{V} \times_2 \mathbf{U}_\mathrm{r} \times_3 \mathbf{U}_\mathrm{c}$. (a) $\mathcal{D}_\theta^\top \mathcal{D}_\theta$ and $\mathcal{R}_\phi^\top \mathcal{R}_\phi$ are diagonal in the orthonormal basis $\mathcal{Q}$, with eigenvalues $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}$ for every band and ν_b for every pixel. (b) The exact data step (6) is

$$\mathbf{Z}_k = \mathcal{Q}\Big(\mathcal{Q}^{-1}(\mathbf{C}_k) \oslash \big[1 + \alpha_k \lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \beta_k \nu_b\big]_{b,i,j}\Big), \tag{10}$$

with $\mathbf{C}_k$ the right-hand side of (6) and $\oslash$ the element-wise division: three mode products in each direction, $O(BHW(B+H+W))$ operations, no iteration and no periodic-border assumption. (c) The null space of the joint operator, $\mathcal{N}(\mathcal{D}_\theta) \cap \mathcal{N}(\mathcal{R}_\phi)$, is spanned by the basis tensors with $\nu_b = 0$ and $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} = 0$; its orthogonal projector is $(\mathbf{I} - \mathbf{P}_\mathrm{s}) \otimes (\mathbf{I} - \mathbf{P}_\mathrm{r} \otimes \mathbf{P}_\mathrm{c})$ in mode order, with $\mathbf{P}_\mathrm{s}$, $\mathbf{P}_\mathrm{r}$, $\mathbf{P}_\mathrm{c}$ the projectors onto the row spaces of $\mathbf{r}_\phi$, $\mathbf{D}_\mathrm{r}$, $\mathbf{D}_\mathrm{c}$. If these have full row rank, its dimension is $(B-M)(HW-hw)$, a fraction $(1-M/B)(1-1/s^2)$ of all unknowns.

Proof: (a) By (9), $\mathcal{D}_\theta^\top \mathcal{D}_\theta$ applies $\mathbf{D}_\mathrm{r}^\top \mathbf{D}_\mathrm{r}$ and $\mathbf{D}_\mathrm{c}^\top \mathbf{D}_\mathrm{c}$ to the two spatial modes and the identity to the spectral mode, and $\mathcal{R}_\phi^\top \mathcal{R}_\phi$ applies $\mathbf{r}_\phi^\top \mathbf{r}_\phi$ to the spectral mode alone; each factor is diagonalized by the corresponding factor of $\mathcal{Q}$, and an identity factor by any orthonormal basis. (b) In this basis the matrix of (6) is diagonal with entries $1 + \alpha_k \lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \beta_k \nu_b > 0$. (c) $\mathcal{N}(\mathcal{D}_\theta) \cap \mathcal{N}(\mathcal{R}_\phi) = \mathcal{N}(\mathcal{D}_\theta^\top \mathcal{D}_\theta + \mathcal{R}_\phi^\top \mathcal{R}_\phi)$, whose eigenvalues $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \nu_b$ are sums of nonnegative terms and vanish only when both terms do. The basis tensors with $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} > 0$ span the row space of $\mathbf{D}_\mathrm{r} \otimes \mathbf{D}_\mathrm{c}$, of dimension hw, and those with $\nu_b > 0$ the row space of $\mathbf{r}_\phi$, of dimension M, which gives the projector and the dimension. ∎

The closest closed form, that of FUSE [7], solves the same system by Fourier transforms under periodic borders; (10) holds for the border rule of the simulation, which Theorem 3 shows to matter, and is a differentiable layer whose operators may change from one image to the next, at the cost of eigendecompositions of an $H \times H$, a $W \times W$ and a $B \times B$ matrix. Part (c) states what any fusion method has to infer from a prior: the high spatial frequencies of the spectral directions that the multispectral sensor does not see, 88% of the unknowns for Chikusei and 90% for Pavia Centre at $\times 4$; every other component is determined by the observations. Section VI-C uses this to split the error of every method into a part that contradicts the observations and a part in this null space.

E. The cost of an inexact operator

Stack the two data terms as $\mathbf{A} = [\mathcal{D}_\theta; \mathcal{R}_\phi]$, $\mathbf{Y} = [\mathbf{Y}_\mathrm{H}; \mathbf{Y}_\mathrm{M}]$ and $\mathbf{W}_k = \mathrm{diag}(\alpha_k \mathbf{I}, \beta_k \mathbf{I})$, and let $\tilde{\mathbf{A}}$ be the operator a network uses in place of $\mathbf{A}$: one with a zero border, a learned one, or one with wrong parameters. Write $T(\mathbf{V}) = (\mathbf{I} + \tilde{\mathbf{A}}^\top \mathbf{W}_k \tilde{\mathbf{A}})^{-1}(\mathbf{V} + \tilde{\mathbf{A}}^\top \mathbf{W}_k \mathbf{Y})$ for the exact data step (6) applied to an estimate $\mathbf{V}$, and $G(\mathbf{V}) = \mathbf{V} - \tilde{\mathbf{A}}^\top \mathbf{W}_k(\tilde{\mathbf{A}}\mathbf{V} - \mathbf{Y})$ for its linearization (7).

Theorem 3 (Fixed point and operator mismatch): Let $\mathbf{Y} = \mathbf{A}\mathbf{X}^*$. (a) If $\tilde{\mathbf{A}} = \mathbf{A}$, then $T(\mathbf{X}^*) = G(\mathbf{X}^*) = \mathbf{X}^*$ for all $\alpha_k, \beta_k > 0$. (b) For every $\tilde{\mathbf{A}}$,

$$\|T(\mathbf{X}^*) - \mathbf{X}^*\| \le \tfrac{1}{2} \|\mathbf{W}_k^{1/2}(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*\|, \tag{11}$$

$$\|G(\mathbf{X}^*) - \mathbf{X}^*\| \le \|\mathbf{W}_k^{1/2}\tilde{\mathbf{A}}\| \, \|\mathbf{W}_k^{1/2}(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*\|. \tag{12}$$

(c) Let stage k of a network be $\mathbf{X}_k = (\mathcal{I} + \mathcal{P}_k)(T_k(\mathbf{X}_{k-1}))$, where the exact data step T_k uses $\mathbf{W}_k$ and the residual prior satisfies $\mathcal{P}_k(\mathbf{X}^*) = \mathbf{0}$ and makes $\mathcal{I} + \mathcal{P}_k$ L_k-Lipschitz. Started at $\mathbf{X}_0 = \mathbf{X}^*$, the network returns

$$\|\mathbf{X}_K - \mathbf{X}^*\| \le \tfrac{1}{2} \sum_{k=1}^{K} \Big(\prod_{j=k}^{K} L_j \Big) \|\mathbf{W}_k^{1/2}(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*\|. \tag{13}$$

Proof: (a) $\tilde{\mathbf{A}}\mathbf{X}^* - \mathbf{Y} = \mathbf{0}$ makes both updates vanish. (b) With $\mathbf{B} = \mathbf{W}_k^{1/2}\tilde{\mathbf{A}}$, a direct computation gives $T(\mathbf{X}^*) - \mathbf{X}^* = (\mathbf{I} + \mathbf{B}^\top \mathbf{B})^{-1}\mathbf{B}^\top \mathbf{W}_k^{1/2}(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*$; in the singular value decomposition of $\mathbf{B}$, $(\mathbf{I} + \mathbf{B}^\top \mathbf{B})^{-1}\mathbf{B}^\top$ has singular values $b_i/(1 + b_i^2) \le \frac{1}{2}$. Similarly $G(\mathbf{X}^*) - \mathbf{X}^* = \mathbf{B}^\top \mathbf{W}_k^{1/2}(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*$, whose norm is at most $\|\mathbf{B}\|$ times the last factor. (c) T_k is nonexpansive, since $\|T_k(\mathbf{V}) - T_k(\mathbf{V}')\| = \|(\mathbf{I} + \mathbf{B}^\top \mathbf{B})^{-1}(\mathbf{V} - \mathbf{V}')\| \le \|\mathbf{V} - \mathbf{V}'\|$. With $e_k = \|\mathbf{X}_k - \mathbf{X}^*\|$ and $(\mathcal{I} + \mathcal{P}_k)(\mathbf{X}^*) = \mathbf{X}^*$, $e_k \le L_k \|T_k(\mathbf{X}_{k-1}) - \mathbf{X}^*\| \le L_k \big(\|T_k(\mathbf{X}_{k-1}) - T_k(\mathbf{X}^*)\| + \|T_k(\mathbf{X}^*) - \mathbf{X}^*\|\big) \le L_k(e_{k-1} + b_k)$, where b_k is the bound (11); unrolling from $e_0 = 0$ gives (13). ∎

Theorem 3 separates the two roles of the operator. With the exact operator, the ground truth is a fixed point of every data step, so the priors only have to remove the reconstruction error. With an inexact one, every data step moves the ground truth, and the priors must also undo this displacement; because it depends on the image, it cannot be removed by a fixed correction. The exact proximal step bounds the displacement by $\frac{1}{2}\|\mathbf{W}_k^{1/2}(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*\|$ whatever the learned weights, whereas for the gradient step the bound grows with $\|\mathbf{W}_k^{1/2}\tilde{\mathbf{A}}\|$, which training may make large, and the step is not nonexpansive, so (13) has no counterpart for it. The theorem also explains why an exact operator can be replaced at test time: for data generated by $\mathbf{A}'$, giving the network $\tilde{\mathbf{A}} = \mathbf{A}'$ restores (a), which a learned operator cannot do.

Corollary 1 (Border rule): If $\tilde{\mathbf{A}}$ differs from $\mathbf{A}$ only in the border rule of $\mathcal{D}_\theta$, e.g., zero instead of reflect padding, $(\mathbf{A} - \tilde{\mathbf{A}})\mathbf{X}^*$ is nonzero only on the LR pixels whose $(2p+1) \times (2p+1)$ kernel support reaches the padding, i.e., within $\lceil p/s \rceil$ pixels of the LR border.

With decimation from index zero and $p < s$, only the first LR row and column are affected, a fraction $(h + w - 1)/(hw)$ of an $h \times w$ LR image: 3% of a 64×64 image but 12% of the 16×16 LR patches on which networks are trained. The displacement (11) is then small, which matches the small accuracy gap at the training blur (Section VI-A), whereas a learned operator differs from $\mathbf{A}$ everywhere. In real images the true border rule is unknown [58], and Corollary 1 bounds what any fixed choice can cost; the main role of exactness is the one stated by Theorem 3: an operator that can be supplied, estimated and replaced. Section VI-H measures the displacement and the test PSNR as functions of the operator residual $\|(\tilde{\mathbf{A}} - \mathbf{A})\mathbf{X}^*\|$.

Algorithm 1 PUFormer inference (unrolled HQS)

Require: $\mathbf{Y}_\mathrm{H}$, $\mathbf{Y}_\mathrm{M}$; operator parameters (θ, ϕ), given or $\arg\min_{\mathcal{G}} J$ of (8)

build the exact $\mathcal{D}_\theta, \mathcal{D}_\theta^\top, \mathcal{R}_\phi, \mathcal{R}_\phi^\top$ (reflect border, (2))
$\mathbf{X}_0 \leftarrow \mathrm{bicubic}_{\times s}(\mathbf{Y}_\mathrm{H})$, $\mathbf{F}_0 \leftarrow \varnothing$
for $k = 1, \dots, K$ **do**
  $\mathbf{Z}_k \leftarrow$ data step: (14), or CG on (6) ▷ fp32
  $\mathbf{r}_k \leftarrow s^2\,\mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_\mathrm{H})$
  $(\Delta_k, \mathbf{F}_k) \leftarrow \mathcal{P}_k([\mathbf{Z}_k, \mathbf{Y}_\mathrm{M}, \mathbf{r}_k], \mathbf{F}_{k-1})$ ▷ fp16
  $\mathbf{X}_k \leftarrow \mathbf{Z}_k + \Delta_k$
end for
return $\mathbf{X}_K$

IV. PUFormer

A. Unfolded iterations

PUFormer (Fig. 1) is the HQS iteration of Section III-B unrolled for $K = 3$ stages, starting from the bicubic upsampling [59] $\mathbf{X}_0$ of $\mathbf{Y}_\mathrm{H}$. Stage k (Fig. 1, bottom) first performs the data step (4) with the exact operators, either linearized,

$$\begin{aligned}\mathbf{Z}_k = \mathbf{X}_{k-1} &- s^2\alpha_k\,\mathcal{D}_\theta^\top(\mathcal{D}_\theta\mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{H}) \\ &- \beta_k\,\mathcal{R}_\phi^\top(\mathcal{R}_\phi\mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{M}),\end{aligned} \tag{14}$$

or exactly, in closed form by (10) (Fig. 2; by conjugate gradients on (6) if the kernel is not separable). The weights $\alpha_k = \mathrm{softplus}(a_k)$ and $\beta_k = \mathrm{softplus}(b_k)$ stand for $1/(\mu_k\varsigma_\mathrm{H}^2)$ and $1/(\mu_k\varsigma_\mathrm{M}^2)$ in (6) and are learned per stage; the factor s^2 compensates the $1/s^2$ gain of $\mathcal{D}^\top\mathcal{D}$ on smooth content under $\times s$ decimation, so that both terms start at a comparable scale. The prior step (5) is a learned residual,

$$\begin{aligned}(\Delta_k, \mathbf{F}_k) &= \mathcal{P}_k\big([\mathbf{Z}_k, \mathbf{Y}_\mathrm{M}, s^2\mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_\mathrm{H})],\, \mathbf{F}_{k-1}\big), \\ \mathbf{X}_k &= \mathbf{Z}_k + \Delta_k,\end{aligned} \tag{15}$$

where the third input, the gradient of the LR-HSI data term at $\mathbf{Z}_k$, tells the prior where the updated estimate still disagrees with $\mathbf{Y}_\mathrm{H}$, and $\mathbf{F}_{k-1}$ are the final decoder features of the previous stage (cross-stage memory, merged by a 1×1 convolution). A variant also gives the prior the back-projected HR-MSI residual $(B/M)\,\mathcal{R}_\phi^\top(\mathcal{R}_\phi\mathbf{Z}_k - \mathbf{Y}_\mathrm{M})$. The last convolution of every prior is initialized to zero, so training starts from the pure HQS iteration with $\mathcal{P}_k = \mathbf{0}$, which leaves the ground truth unchanged as required by Theorem 3(c). The image path, including all operator applications, stays in 32-bit floating point; only the priors run in 16-bit mixed precision [60].

Learned spectral data step. In the basis of Theorem 2, the exact step multiplies every component of the data residual by the gain $1/(1+\alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}+\beta_k\nu_b)$, the filter of a white Gaussian prior. Natural images are not white, and noise does not harm all components equally, so we let two small networks modulate these gains:

$$\begin{aligned}\mathbf{Z}_k &= \mathbf{X}_{k-1} + \mathcal{Q}\big(\mathcal{Q}^{-1}(\mathbf{G}_k) \odot \mathbf{\Gamma}_k\big), \\ \Gamma_{k,bij} &= \frac{\exp\big(\ln 2\,\tanh(u_k(\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}, \mathbf{c}) + v_k(\nu_b, \mathbf{c}))\big)}{1 + \alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \beta_k\nu_b},\end{aligned} \tag{16}$$

where $\mathbf{G}_k = \alpha_k\mathcal{D}_\theta^\top(\mathbf{Y}_\mathrm{H} - \mathcal{D}_\theta\mathbf{X}_{k-1}) + \beta_k\mathcal{R}_\phi^\top(\mathbf{Y}_\mathrm{M} - \mathcal{R}_\phi\mathbf{X}_{k-1})$ is the data residual and u_k, v_k are two-layer perceptrons of the log-eigenvalue and of the observation code $\mathbf{c}$ introduced below, initialized to zero so that training starts from the exact step (10). The gain is separable, so a stage evaluates the networks $HW + B$ times rather than BHW times.

Proposition 1: For all u_k, v_k: (a) with exact operators and noise-free data, $\mathbf{X}^*$ is a fixed point of (16); (b) (16) is nonexpansive; (c) with an inexact operator $\tilde{\mathbf{A}}$, it moves $\mathbf{X}^*$ by at most $\|\mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*\|$, twice the bound (11) and independent of the weights.

Proof: Let $\varsigma_{bij} = \alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \beta_k\nu_b$, the eigenvalues of $\tilde{\mathbf{A}}^\top\mathbf{W}_k\tilde{\mathbf{A}}$, and e^f the numerator of Γ, so $\frac{1}{2} < e^f < 2$. (a) $\mathbf{G}_k = \mathbf{0}$ at $\mathbf{X}^*$. (b) The step maps a difference $\mathbf{V} - \mathbf{V}'$ to $\mathcal{Q}\big((1 - \Gamma\varsigma) \odot \mathcal{Q}^{-1}(\mathbf{V} - \mathbf{V}')\big)$, and $0 \le \Gamma\varsigma = e^f\varsigma/(1+\varsigma) < 2$, so every factor lies in $(-1, 1]$. (c) The displacement is $\mathcal{Q}\big(\Gamma\odot\mathcal{Q}^{-1}(\mathbf{B}^\top\mathbf{e})\big)$ with $\mathbf{B} = \mathbf{W}_k^{1/2}\tilde{\mathbf{A}}$ and $\mathbf{e} = \mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*$; its operator norm is $\max e^f\sqrt{\varsigma}/(1+\varsigma) \le 2\cdot\frac{1}{2} = 1$. ∎

Theorem 3(a) and (c) therefore hold for the learned step too: it can reweight how much the network trusts each observed component, but it cannot move a correct estimate or amplify errors.

B. Transposed-attention prior

Each prior (Fig. 3) is a three-level U-Net [61] with widths 48, 96 and 192, two, three and four transformer blocks on the encoder levels and bottleneck, mirrored decoder levels, and two refinement blocks. A block (Fig. 4(a)) follows Restormer [44]: multi-Dconv-head transposed attention (Fig. 5) computes queries, keys and values with 1×1 and depth-wise 3×3 convolutions, normalizes each feature map over space, and forms a $c' \times c'$ attention matrix across channels per head (1, 2 and 4 heads on the three levels) with a learned temperature. Attention is thus computed between spectral feature maps rather than between spatial positions, its cost is linear in the number of pixels, and the receptive field of every block covers the whole image. A gated depth-wise feed-forward network with expansion 2.66 follows (Fig. 4(b)). The three priors hold 3.37–3.38 M parameters each; the whole network has 10.13 M parameters for Chikusei and 9.99 M for Pavia Centre, whose input and output convolutions are smaller.

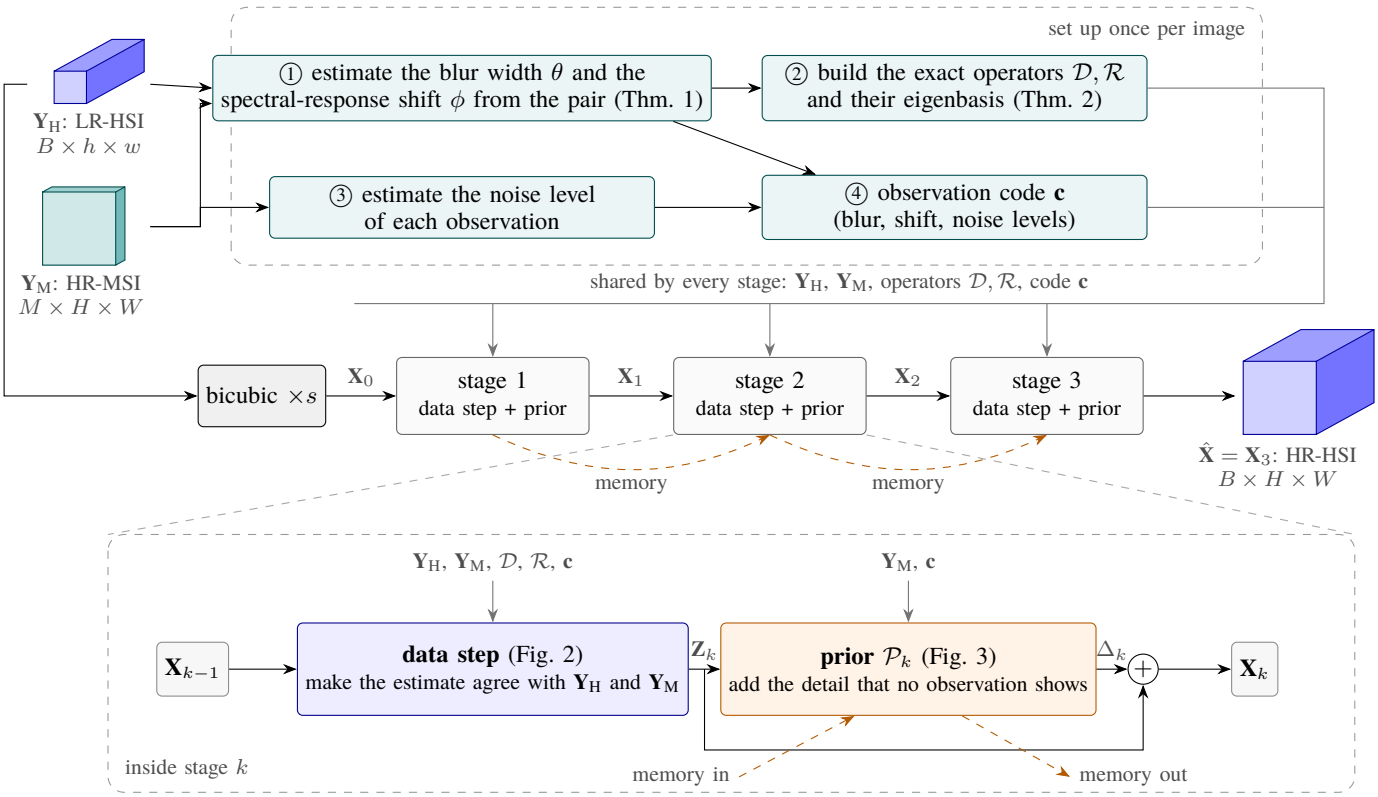


Fig. 1. PUFormer, the HQS iteration of Section III-B unrolled for three stages. Top: the observation model is set up once per image: the operator parameters are given or estimated from the observation pair (Theorem 1), the exact operators and their eigenbasis are built (Theorem 2), and the noise levels form the observation code $\mathbf{c}$. Middle: every stage refines the estimate and passes its decoder features to the next (memory). Bottom: inside a stage, a data step makes the estimate agree with both observations, and a learned prior adds the detail that the observations cannot provide.

C. Training objective

With ground truth $\mathbf{X}$, the loss is

$$\mathcal{L} = \|\mathbf{X}_K - \mathbf{X}\|_1 + 0.1 \sum_{k<K} \|\mathbf{X}_k - \mathbf{X}\|_1 + \lambda_{\mathrm{SAM}}\,\mathrm{SAM}(\mathbf{X}_K, \mathbf{X}) + \lambda_{\mathrm{S}}\big(1 - \mathrm{SSIM}(\mathbf{X}_K, \mathbf{X})\big), \tag{17}$$

with mean absolute errors, deep supervision [62] of the intermediate stages, the mean spectral angle in radians, and a band-wise Gaussian SSIM [63] whose data range is the maximum of each training patch. We use $\lambda_{\mathrm{SAM}} = 0.05$ and $\lambda_{\mathrm{S}} = 0.1$; both terms are computed in 32-bit precision. An exponential moving average of the weights (decay 0.999) [64] is used for validation and testing.

D. Training for a family of observation models

Trained with one fixed operator pair on noise-free data, the priors adapt to one blur, one spectral response and no noise, and Section VI-G shows the consequences: a shifted spectral response or noise is not compensated by giving the network the true operator. Because the operators enter PUFormer only through its data steps, the same network can instead be trained for a family of observation models. For every batch we draw a blur width $\sigma \sim \mathcal{U}[0.5, 2]\,\sigma_0$, a shift $\delta \sim \mathcal{U}[-12, 12]$ nm of the spectral responses and, with probability 0.5, Gaussian noise on both observations at an SNR drawn from $\mathcal{U}[30, 50]$ dB; the batch is simulated with $(\mathcal{D}_\sigma, \mathcal{R}_\delta)$, and the same operators are used in the data steps. At test time the operators are given, or estimated from the test pair by minimizing (8). Training on randomized degradations is a standard remedy in blind super-resolution [65], [66], and USRNet [45] passes the blur kernel to the data step of an unfolded single-image super-resolution network. We extend this to the two operators of fusion, with exact borders and with both operators estimated from the observation pair, and compare it with randomized training alone, in which the data steps keep the nominal operator.

Observation-model conditioning. The data steps then know the operator, but the prior does not, and neither knows the noise level, although the HQS derivation (Section III-B) shows that the weights α_k, β_k should follow the noise of the two observations and that the proximal step depends on it. We therefore describe each input pair by a code $\mathbf{c} = (\sigma/\sigma_0 - 1,\ \delta/12,\ \log_{10} n_\mathrm{H} + 2,\ \log_{10} n_\mathrm{M} + 2)$, where (σ, δ) are the parameters of the operator used in the data steps and n is the noise standard deviation of an observation relative to its RMS, estimated from the observation itself as the median absolute diagonal second difference over all bands [67]. The step sizes become $\alpha_k = \mathrm{softplus}(a_k + \mathbf{u}_k^\top \mathbf{c})$ and $\beta_k = \mathrm{softplus}(b_k + \mathbf{v}_k^\top \mathbf{c})$, so that the two data terms can be weighted by their noise as in a maximum a posteriori estimate, and an embedding of $\mathbf{c}$ (two linear layers) scales and shifts the features of every prior (FiLM [68]) after its input convolution, at the input of its bottleneck and before its output convolution. All added parameters start at zero, so training starts from the

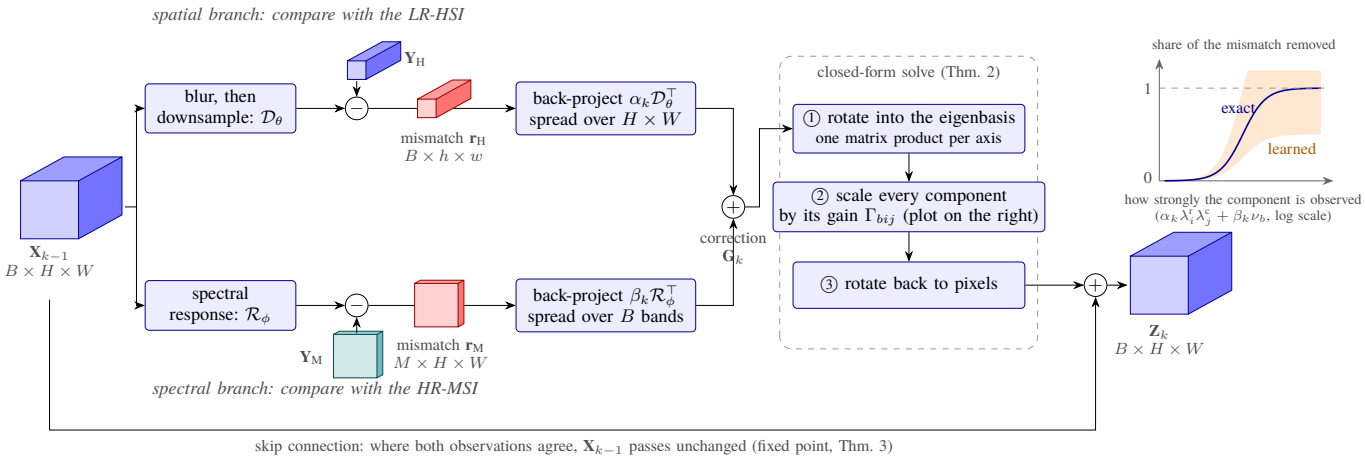


Fig. 2. The data step of stage k, the exact solution of (6). Two physics branches compare the estimate with the observations: the spatial branch blurs and downsamples it and compares it with the LR-HSI, the spectral branch applies the multispectral responses and compares it with the HR-MSI; both mismatches are spread back to the full cube by the exact adjoints. The closed-form solve rotates this correction into the eigenbasis of the separable operators, removes in each component a share $\Gamma_{bij}(\alpha_k \lambda_i^\mathrm{r} \lambda_j^\mathrm{c} + \beta_k \nu_b)$ of the mismatch that grows with how strongly the observations see that component (blue: the exact step (10); orange: the range allowed to the learned gains of Proposition 1), and rotates back. Components that neither observation sees (the null space) are left to the prior.

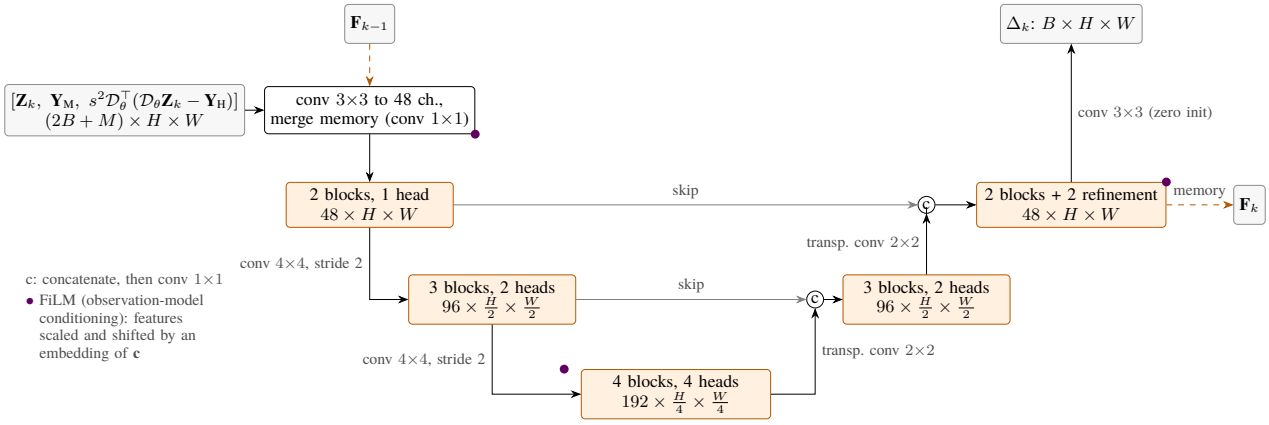


Fig. 3. The prior $\mathcal{P}_k$ of stage k (widths for Chikusei, $B = 128$, $M = 8$): a three-level U-Net of transformer blocks (Fig. 4). The encoder halves the resolution twice with strided convolutions, the decoder restores it with transposed convolutions and merges the encoder features through skip connections; the previous stage's decoder features enter after the input convolution, and the last convolution starts at zero so that training starts from the HQS iteration.

unconditioned network; at the nominal operator without noise, $\mathbf{c}$ is nearly constant and the network reduces to PUFormer.

E. Self-ensemble

At test time we average the prediction for the input with the transposed prediction for the transposed input [69]. Transposition preserves the decimation phase, so both passes see observations generated by exactly the same operator. Flips and rotations do not: flipping a 256-pixel row maps sampled position $4m$ to $255 - 4m$, which has phase three, so we do not use them.

V. Experimental Setup

A. Data and protocol

Chikusei [70] is an airborne scene of 2517×2335 pixels with 128 bands from 363 to 1018 nm. The split sizes follow [15]: eight 256×256 test images, 64 validation patches and 832 training patches of 64×64. Because their locations are not specified there, we fix them as follows. We take the central 2048×2048 crop, which contains no no-data border, and divide it by its global maximum (15,133 digital numbers, DN). The test images are the eight 256×256 tiles of the top 256 rows, the validation patches tile rows 272–400, and training draws random 64×64 crops with dihedral augmentation from rows 416–2048 (800 non-overlapping patch equivalents). Sixteen-row gaps keep the three regions disjoint.

Pavia Centre [71] was recorded by the ROSIS sensor over the centre of Pavia, Italy; the distributed scene has 1096×715 pixels at 1.3 m and 102 bands between 430 and 860 nm. For this dataset, [15] uses the benchmark files released with PSRT [10]: 92 bands, a four-band multispectral image, four 256×256 test images and 142 validation and 1284 training patches of 64×64. These files are distributed through a file-sharing service that we could not access, so we rebuild the benchmark from the original scene to the same specification. We keep bands 11–102 (473–860 nm if the 102 bands are taken

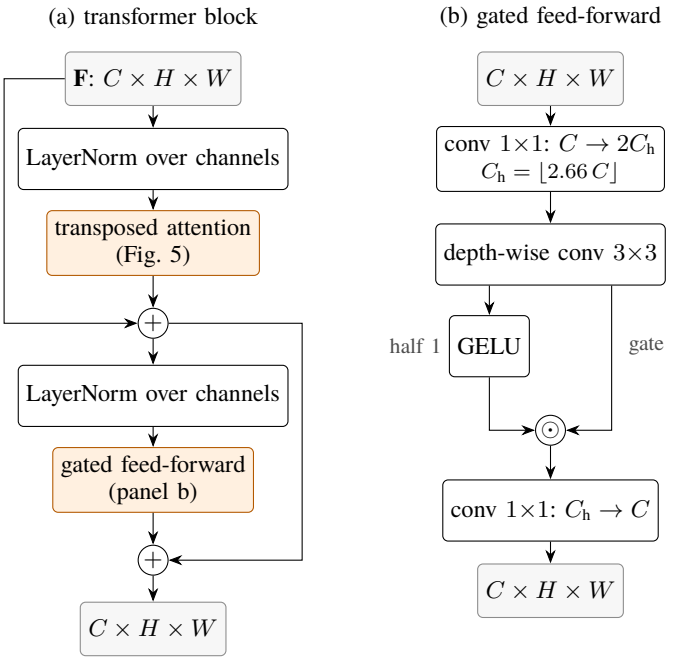


Fig. 4. (a) A transformer block of the prior: layer normalization over channels, transposed attention and a gated feed-forward network, each in a residual branch. (b) The gated feed-forward network: the expanded features are split in two halves, and the GELU of one half gates the other.

as evenly spaced over 430–860 nm), divide by the maximum of the retained bands (8,000 DN), apply the same blur and decimation as for Chikusei, and simulate the HR-MSI with the IKONOS responses. The test images are the four 256×256 tiles of the top-left 512×512 region. The 63 validation patches lie on a 32-pixel grid in rows 528–656, and training draws random 64×64 crops with dihedral augmentation from rows 672–1096 and from the strip to the right of the test region (rows 0–512, columns 528–715), 97 non-overlapping patch equivalents in total. Gaps of 16 pixels keep the regions disjoint. As a check on the rebuilt data, our GSA implementation reaches 34.78 dB on our test images, against 35.35 dB reported for GSA in [15].

For both datasets we follow Wald's protocol [72]: the original image serves as ground truth and both observations are simulated from it, and every image, including every training crop, is degraded on its own with (1). Table I summarizes the two protocols.

Equal-budget benchmark. Reported numbers come from different splits, normalizations and metric code (Section V-B), so for the main comparison we retrain every learned method ourselves under one pipeline: SSR-NET [8], PSRT [10], DCT [54], MIMFormer [26] and DHIF-Net [20], built from their authors' code[1]. They cover a CNN, a pyramid window transformer, the dual-cross transformer that is among the strongest entries of [15], a multiscale mixer, and a model-guided unfolding network whose blur, decimation and spectral operators are learned convolution layers. Every method, PUFormer included, is trained for the same 4 h on one Tesla T4, with the same random patches and augmentation, batch size, EMA weights and validation-only model selection. The published networks keep their official optimizer settings (Adam, L1 loss, learning rate 10^{-4}, or 3×10^{-4} for DHIF-Net, no weight decay or gradient clipping) and are trained in 32-bit precision as released; DCT and MIMFormer use gradient checkpointing [73] to fit into the 16 GB of a T4, which does not change what they compute. The test metrics are computed by the same code for all methods.

[1]PSRT: github.com/Deng-shangqi/PSRT (commit 9c2be90); DCT: github.com/qingma2016/DCTransformer (bd75b15); MIMFormer: github.com/meiruni/MIMFormer (0904158); DHIF-Net: github.com/TaoHuang95/DHIF-Net (6434366). Only the numbers of input and output bands are changed; DHIF-Net's code covers $\times8$ and $\times16$, and at $\times4$ we use one of the two stride-4 layers of its $\times16$ operator.

TABLE I
THE TWO $\times4$ PROTOCOLS (DEGRADATION AND SPLIT SIZES OF [15]; IMAGE LOCATIONS OURS)

Item	Chikusei	Pavia Centre
HR-HSI	central 2048^2 crop, 128 bands	1096×715, bands 11–102 (92)
Normalization	$\div$ 15,133 DN	$\div$ 8,000 DN
LR-HSI	Gaussian 7×7, $\sigma = 2$; $\times4$ decimation; reflect border	
HR-MSI	WorldView-2, 8 bands	IKONOS, 4 bands
Test	8 of 256^2 (rows 0–256)	4 of 256^2 (rows, columns 0–512)
Validation	64 of 64^2 (rows 272–400)	63 of 64^2 (rows 528–656)
Training	random 64^2 crops, rows 416–2048	random 64^2 crops, see text
Selection	validation PSNR only	

B. Metrics

We report the eight metrics of [15], averaged over the test images: PSNR; SSIM [63]; the spectral angle mapper (SAM, degrees, averaged over the pixels whose ground-truth and estimated spectra are both nonzero) [74]; ERGAS with ratio 4 [75]; Q2n [76] on 32×32 blocks, computed on the image in DN exactly as in the reference implementation of the pan-sharpening toolbox [77], [78]; the band-averaged correlation coefficient (CC); the spatial correlation coefficient (SCC) [79] of Sobel gradient magnitudes as in the same toolbox; and RMSE in DN. The headline PSNR is $10\log_{10}(1/\mathrm{MSE})$ over the whole normalized image, i.e., its peak is the dataset maximum. The PSNR values of [15] also use a fixed peak: for every row of their Chikusei table, $\mathrm{RMSE}\cdot10^{\mathrm{PSNR}/20}$ is between 16.6k and 17.1k DN. Since our peak is lower (15,133 DN), our headline PSNR is conservative; with a 16.7k DN peak it would be 0.86 dB higher. For Pavia Centre the same product lies between 7.7k and 8.4k DN (median 8.07k), close to our peak of 8,000 DN, so there the PSNR values are on the same scale. The SSIM definition of [15] is not stated. We report SSIM as in the code released with PSRT [10], whose benchmark data [15] adopts (Gaussian 11×11 window, $\sigma = 1.5$, data range 1, i.e., the dataset peak), and additionally a stricter variant whose data range is each image's own maximum. We also report observation consistency, $\mathrm{PSNR}(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{H})$ and $\mathrm{PSNR}(\mathcal{R}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{M})$.

C. Implementation

PUFormer is implemented in PyTorch [80] and trained on Kaggle NVIDIA Tesla T4 GPUs (Table II). On Chikusei we train in two runs. The first run trains from scratch on one GPU with zero-padded operators and the L1 losses only. The final model fine-tunes it with the exact reflect-border operators

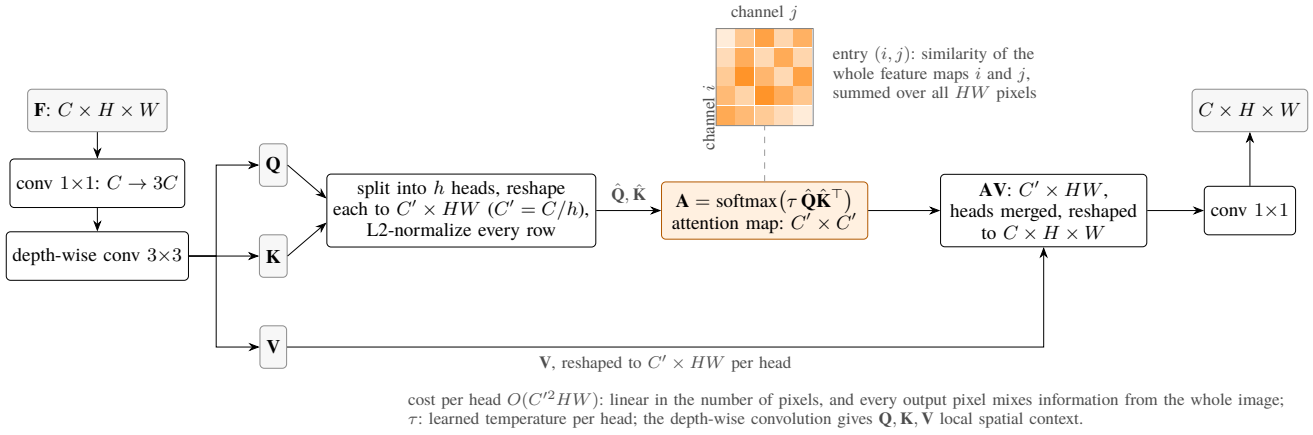


Fig. 5. Multi-Dconv-head transposed attention [44], the token mixer of the prior. Queries, keys and values are computed by point-wise and depth-wise convolutions, and the attention map is formed between the C' feature channels of each head rather than between the HW pixels, so its size does not depend on the image size. Replacing it by a depth-wise convolution changes the PSNR by less than 0.1 dB in the ablation.

TABLE II
TRAINING RUNS

	Chikusei		Pavia Centre
	First run	Fine-tuned (final)	final
Initialization	random	first run	random
Border of $\mathcal{D}$	zero padding	reflect (exact)	reflect (exact)
Loss	L1 + 0.1 deep sup.	(17)	(17)
GPUs × batch	1×8	2×8	2×8
Peak learning rate	3×10^{-4}	1.5×10^{-4}	3×10^{-4}
Iterations	88,946	94,450	93,850
Training patches	0.71 M	1.51 M	1.50 M
Wall-clock time	9.5 h	10.9 h	10.2 h
Selected at	best val. (58.60 dB)	it. 44,000 (58.72 dB)	it. 43,000 (48.58 dB)

and the full loss (17) on two GPUs with distributed data parallelism [81]. On Pavia Centre, a single run trains the model from scratch with the exact operators and the full loss on two GPUs, within one 12-hour session that also runs all tests. All runs use AdamW [82] ($\beta = (0.9, 0.99)$, weight decay 10^{-4}), a 1000-iteration linear warm-up followed by a cosine decay [83] to 10^{-3} of the peak rate that ends at the time budget, and gradient-norm clipping at 0.1. The weights with the best validation PSNR are tested once. The ablation and benchmark runs (Sections VI-A and VI-B) train from scratch with these settings, the full loss and a single GPU for 4 h each; their models are evaluated without self-ensemble.

VI. Results

A. Ablation study

Table III changes one module of the full model at a time and retrains each variant from scratch with the same budget (4 h on one T4, about 40,000 iterations for the full model). On Pavia Centre the full model and the zero-border variant were trained with three seeds; their PSNR varies by only 0.02 and 0.03 dB across seeds, which sets the scale for the other differences. Chikusei repeats every variant, again with three seeds for these two rows.

Data step and operator. Removing the data-consistency step together with the residual input (row (2)) costs 0.17 dB and lowers the agreement of the reconstruction with the LR-HSI from 77.6 to 68.9 dB. Keeping the exact kernel but replacing the reflect border by zero padding (row (4)) costs 0.08 dB on average over the three seeds, and each zero-border seed is below each full-model seed. A blur kernel and spectral response learned with the network (row (5)), the operator model of DHIF-Net and MHF-Net, cost 0.06 dB. Both variants reproduce the LR-HSI about 3 dB worse than the exact model (74.5 and 73.6 against 77.6 dB). This is what Theorem 3 predicts: their data steps move the ground truth, and the prior has to compensate. The compensation is tied to the operator used in training: Chikusei weights trained with a zero border fall from 58.60 to 49.52 dB validation PSNR when the exact border is substituted without retraining (Section S-I of the supplementary material). On the test images, the displacement of the ground truth by one LR-HSI data step, $\|\tilde{\mathcal{D}}^\top(\tilde{\mathcal{D}} - \mathcal{D})\mathbf{X}\|$ relative to $\|\tilde{\mathcal{D}}^\top\tilde{\mathcal{D}}\mathbf{X}\|$, is 4.66% for the zero border and 4.69% for the learned operator, and on 64×64 crops, where border pixels are a larger share, 11.11% and 11.17%. Chikusei (rows (2)–(17)) shows the same ordering with larger gaps: without the data step the PSNR is 0.30 dB lower, with a zero border 0.22 dB and with a learned operator 0.22 dB, and the agreement with the LR-HSI drops from 83.9 to 73.3, 75.9 and 77.1 dB.

The accuracy gap at the training blur is small; it becomes large when the blur changes (last column of Table III and Fig. 6). Given the true blur at test time, the exact model loses at most 0.13 dB for $\sigma \in \{1.5, 2.5, 3.0\}$, the zero-border model with the same swap 0.88 dB, and the learned operator, which cannot be replaced, 1.24 dB. Without the swap the exact model loses 7.50 dB: it is the most sensitive variant to a wrong operator, and the only one that fully recovers when given the right one. On Chikusei the difference is larger still: with the true blur the exact model loses at most 0.25 dB, the zero-border model 3.41 dB and the learned operator 3.45 dB. A zero border thus prevents the swap from working, even though it differs from the exact operator only at the image border, because the prior has learned to compensate a border error tied to the training blur.

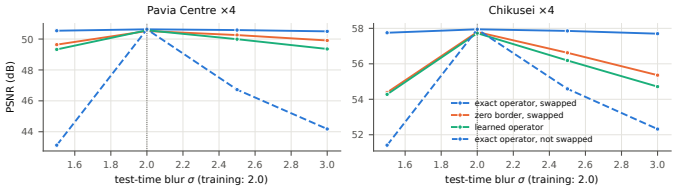


Fig. 6. Test PSNR when the blur of the test data differs from the training blur ($\sigma = 2$, dotted line), for the exact operator with and without the true blur supplied at test time, the zero-border operator with the true blur supplied, and the learned operator, which cannot be replaced.

Prior. Cross-stage memory (row (10)), the residual input (row (3)), the type of token mixer (rows (14) and (15): a depth-wise convolution [84] or self-attention within 8×8 windows [85] instead of transposed attention) and the number of stages (rows (11)–(13)) change the PSNR on Pavia Centre by at most 0.06 dB, comparable to the spread across seeds. A one-stage model with 3.33 M parameters is within 0.02 dB of the full model with 9.99 M; part of this parity comes from the fixed time budget, in which the smaller model completes almost three times as many iterations. The SAM and SSIM loss terms (row (16)) do not improve the PSNR, whereas on Pavia Centre deep supervision of the intermediate stages does (row (17); 0.12 dB lower without it). On Chikusei, where the three seeds of the full model differ by only 0.004 dB, the same modules change the PSNR by between -0.02 and +0.07 dB: memory +0.01, residual input +0.06, window attention +0.03, convolutional mixer +0.07, two and four stages -0.02 and +0.01, SAM and SSIM loss terms +0.02 and deep supervision +0.04 dB. Several of these simplifications are slightly better than the full model, so its prior is not optimal for this dataset, and deep supervision, which helps on Pavia Centre, does not help here. Only a single stage costs noticeably (0.20 dB): on the larger, 128-band problem at least two stages are needed. We keep the full model for all other experiments, but the ablation shows that PUFormer's accuracy depends more on its exact data-consistency steps than on the particular design of its transformer prior.

B. Equal-budget benchmark

Table IV compares PUFormer with the five retrained networks; CC and SCC are listed in Table S-VI of the supplementary material. On Chikusei PUFormer reaches 57.95 dB, 0.99 dB above the best retrained network, and on Pavia Centre 50.63 dB (+1.00 dB), with 10% and 11% lower RMSE; it also has the lowest SAM and ERGAS on both datasets. MIMFormer is the strongest competitor on both. The unfolded DHIF-Net, whose operators are learned convolutions, reaches 54.39 and 47.67 dB; with 3,250 and 3,950 iterations in 4 h it is also the slowest network to train (Table XII).

Two checks address the fairness of a fixed budget. First, the retrained baselines reach or exceed the values reported for them in [15] although our data and splits differ: on Chikusei $\times 4$, for example, MIMFormer reaches 56.96 dB against 53.36 dB reported and PSRT 53.99 dB against 50.67 dB, so our pipeline does not weaken them. Second, the budget favors fast networks: DCT completes only 4,800 iterations in 4 h on Pavia Centre, against 37,450 for PUFormer and 97,400 for MIMFormer. Doubling the budget to 8 h for PUFormer and the strongest competitor, MIMFormer, gives 50.67 and 49.97 dB on Pavia Centre $\times 4$: the gap narrows from 1.00 to 0.69 dB but remains. The fixed budget also explains why the small SSR-NET, which completes its 300,000 iterations in about 1.5 h, ranks third on both datasets, ahead of the slower PSRT, DHIF-Net and DCT. On Chikusei $\times 4$, PUFormer is more accurate than the best retrained network on all test images (one-sided Wilcoxon signed-rank test [86], $p = 0.0039$). Fig. 7 shows the error of every method on the test image of each dataset on which PUFormer is least accurate. GSA's error follows every edge; the errors of PSRT, DCT and DHIF-Net concentrate on the built-up areas and along building edges, MIMFormer's are smaller, and PUFormer's are the smallest and the most uniform.

Where the margin comes from. Table V retrains the two strongest baselines with their authors' own recipes and reduces PUFormer step by step. The authors' recipes do not help under the common budget: MIMFormer reaches 56.00 instead of 56.96 dB on Chikusei and PSRT 49.51 instead of 53.99 dB, so the common pipeline does not put them at a disadvantage. Reduced to its prior alone, a single U-Net on the bicubic image and the HR-MSI without data step or stages, PUFormer still reaches 57.67 dB on Chikusei and 50.39 dB on Pavia Centre, above every retrained network. Most of the margin over them therefore comes from this residual multiscale prior and its training, and the unrolled exact data steps add 0.28 and 0.24 dB on top; the error split of Section VI-C attributes the margin in the same way.

C. Where the error lies

Theorem 2(c) splits the error of any fused image into a component in the observed subspace $\mathrm{row}(\mathcal{D}) + \mathrm{row}(\mathcal{R})$, which contradicts the observations, and a component in the fusion null space, which no data step can correct and only a prior can reduce. Table VI applies this split to the test predictions of every method of the benchmark. Bicubic interpolation and GSA err mostly in the observed subspace: only 8.6% and 13.7% of their squared error lies in the null space on Chikusei, so they do not even reproduce their inputs. The learned methods differ mainly in how much of the observed part they get wrong. PUFormer's error in the observed subspace is 8.6 times smaller than MIMFormer's on Chikusei and 5.8 times on Pavia Centre, and 98.5% of its remaining error lies in the null space. The split also attributes PUFormer's margin: correcting MIMFormer's observed part alone would raise its PSNR by 0.43 dB on Chikusei, and the remaining 0.62 dB come from a better estimate of the null-space component. The exact data steps account for the first part of the margin and the prior for the second; neither the attention mechanism nor any other single module of the prior is responsible for it (Section VI-A).

D. Other datasets and scale factors

Table VII repeats the equal-budget benchmark on the indoor CAVE [87] and Harvard [88] datasets at $\times 4$, whose test scenes are unseen during training, and at $\times 8$ and $\times 16$, with the blur

TABLE III

ABLATION: EACH VARIANT CHANGES ONE PART OF THE FULL MODEL AND IS TRAINED FROM SCRATCH WITH THE SAME 4 H BUDGET ON ONE T4 (TEST SET, SINGLE PASS). Δ: PSNR CHANGE AGAINST THE FULL MODEL. CONS.: PSNR($\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_{\mathrm{H}}$) IN DB. SWAP: WORST TEST PSNR OVER THE BLURS $\sigma \in \{0.75, 1.25, 1.5\}\sigma_0$ WITH THE TRUE BLUR GIVEN AT TEST TIME; N/A: NO OPERATOR TO REPLACE. $\pm$: MEAN AND STANDARD DEVIATION OVER THREE SEEDS.

			Pavia Centre $\times 4$				Chikusei $\times 4$			
#	Variant	Params (M)	PSNR	Δ	Cons.	Swap	PSNR	Δ	Cons.	Swap
(1)	PUFormer (full model)	9.99	50.62±0.02	–	77.6	50.50	57.95±0.004	–	83.9	57.70
(2)	no data step, no residual input	9.99	50.46	-0.17	68.9	n/a	57.65	-0.30	73.3	n/a
(3)	no residual input	9.99	50.62	-0.01	75.7	–	58.01	+0.06	82.8	–
(4)	exact kernel, zero border	9.99	50.54±0.03	-0.08	74.5	49.64	57.73±0.05	-0.22	75.9	54.38
(5)	learned blur kernel and SRF	9.99	50.57	-0.06	73.6	n/a	57.73	-0.22	77.1	n/a
(6)	exact proximal step, conjugate gradients	9.99	50.55	-0.07	79.4	50.44	57.74	-0.21	85.2	57.54
(7)	exact proximal step, closed form	9.99	50.65	+0.02	80.9	50.58	57.93	-0.02	87.3	57.76
(8)	learned spectral data step	9.99	50.63	+0.01	82.9	50.50	58.00	+0.05	89.4	57.83
(9)	with the HR-MSI residual input	10.11	50.64	+0.01	79.0	50.58	58.04	+0.09	86.5	57.85
(10)	no cross-stage memory	9.98	50.67	+0.05	77.2	–	57.96	+0.01	84.1	–
(11)	$K = 1$ stage	3.33	50.60	-0.02	71.3	–	57.75	-0.20	77.4	–
(12)	$K = 2$ stages	6.66	50.64	+0.02	75.0	–	57.92	-0.02	81.0	–
(13)	$K = 4$ stages	13.32	50.68	+0.06	80.2	–	57.96	+0.01	85.0	–
(14)	convolutional mixer	9.94	50.60	-0.02	78.5	–	58.02	+0.07	85.2	–
(15)	8×8 window attention	9.99	50.63	+0.01	78.6	–	57.98	+0.03	84.4	–
(16)	L1 + deep supervision only	9.99	50.66	+0.03	77.5	–	57.97	+0.02	82.2	–
(17)	no deep supervision	9.99	50.50	-0.12	77.7	–	57.99	+0.04	84.1	–

TABLE IV

EQUAL-BUDGET BENCHMARK: EVERY LEARNED METHOD RETRAINED BY US UNDER THE IDENTICAL PROTOCOL AND BUDGET (4 H ON ONE T4, THE SAME DATA, PATCHES, BATCH, EMA WEIGHTS AND VALIDATION-ONLY MODEL SELECTION; SINGLE PASS). GSA AND BICUBIC INTERPOLATION NEED NO TRAINING. BEST LEARNED METHOD IN BOLD; SSIM WITH THE DATA RANGE SET TO EACH IMAGE'S MAXIMUM; RMSE IN DN.

	Chikusei $\times 4$						Pavia Centre $\times 4$					
Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	RMSE↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	RMSE↓
Bicubic	33.63	0.7404	4.406	9.011	0.6914	321.81	25.52	0.6282	6.554	9.852	0.6806	436.32
GSA [4]	43.20	0.9778	2.730	3.279	0.9596	107.69	34.78	0.9438	5.229	3.494	0.9276	157.93
SSR-NET [8]	55.55	0.9958	0.954	1.707	0.9901	25.30	48.45	0.9954	2.443	0.758	0.9719	30.46
PSRT [10]	53.99	0.9948	1.057	1.794	0.9887	30.57	47.87	0.9958	2.212	0.801	0.9716	32.60
DCT [54]	53.18	0.9945	1.133	2.066	0.9871	33.58	46.94	0.9953	2.442	0.888	0.9712	36.26
MIMFormer [26]	56.96	0.9964	0.801	1.488	0.9918	21.65	49.63	0.9967	1.982	0.674	0.9738	26.66
DHIF-Net [20]	54.39	0.9953	1.020	1.660	0.9900	29.37	47.67	0.9956	2.295	0.840	0.9704	33.42
PUFormer	**57.95**	**0.9969**	**0.731**	**1.344**	**0.9929**	**19.45**	**50.63**	**0.9970**	**1.897**	**0.621**	**0.9754**	**23.68**

scaled to a $(2s-1) \times (2s-1)$ Gaussian with $\sigma = s/2$; all metrics are in the supplementary material. In the 2 settings in which every network has run, PUFormer has the highest PSNR in 2 and the lowest SAM in 2, with PSNR margins over the best retrained network between 0.29 and 1.64 dB. Table VIII repeats the operator ablation in these settings: the exact operator with the true blur supplied loses at most 0.28 dB over the changed test blurs, against 0.3–3.4 dB for the zero-border and learned operators.

E. Comparison with reported results on Chikusei

The comparisons in this and the next subsection place PUFormer against the 17 methods tabulated in [15]. They are less controlled than Section VI-B: the other rows are copied from [15], our test images follow its protocol but are not necessarily the same pixels, and the PUFormer models here were trained longer (Table II). Table IX lists all Chikusei results of [15] together with PUFormer. The compared methods are GSA [4], CNMF [5], FUSE [7], U2Net [34], PSRT [10], DCT [54], DDIF [14], MIMFormer [26], MIMO-SST [11], OTPNet [35], SMGU [39], AMSF [27], SSDT [13], ASMNet [28], DPFormer [29], CLSNet [53] and the two-stage conditional diffusion model [15]; their values are copied from [15].

PUFormer ranks first of 18 on PSNR, SSIM (PSRT definition), SAM, ERGAS, CC, SCC and RMSE, with or without the self-ensemble. Relative to the best reported method, PSNR improves by 2.01 dB, RMSE by 26.4%, ERGAS by 12.2% and SAM by 1.1%. It ranks tenth on Q2n. Section S-III of the supplementary material shows that the Q2n deficit

TABLE V
WHERE THE MARGIN OVER THE RETRAINED NETWORKS COMES FROM (4 H ON ONE T4, SINGLE PASS). THE TWO STRONGEST BASELINES ARE RETRAINED WITH THEIR AUTHORS' OWN RECIPE AND WITH PUFORMER'S; PUFORMER IS REDUCED TO ITS PRIOR ALONE, TO ONE STAGE, TO THREE STAGES WITHOUT THE DATA STEP, AND TRAINED WITH THE BASELINES' RECIPE. Δ: PSNR AGAINST THE FULL PUFORMER OF THE SAME DATASET; IT.: TRAINING ITERATIONS COMPLETED.

	Chikusei ×4				Pavia Centre ×4			
Model and training	PSNR	Δ	SAM	it.	PSNR	Δ	SAM	it.
MIMFormer								
pipeline recipe (batch 8, AdamW, no decay, cosine)	56.96	-0.99	0.801	100k	49.63	-1.00	1.982	97k
authors' recipe (batch 16, Adam, step decay)	56.00	-1.95	0.874	68k	49.32	-1.31	2.022	68k
PUFormer's recipe (lr 3×10^{-4}, decay, clipping, SAM+SSIM)	–	–	–	–	–	–	–	–
PSRT								
pipeline recipe (batch 8)	53.99	-3.95	1.057	64k	47.87	-2.76	2.212	65k
authors' recipe (batch 32, AdamW decay, clipping, constant lr)	49.51	-8.44	1.551	17k	45.44	-5.19	2.597	17k
PUFormer								
prior alone: one U-Net, no data step	57.67	-0.28	0.747	113k	50.39	-0.24	1.934	115k
one stage with data step	57.75	-0.20	0.743	97k	50.60	-0.03	1.909	109k
three stages, no data step	57.65	-0.30	0.752	41k	50.46	-0.17	1.928	42k
baselines' recipe (lr 10^{-4}, no decay, no clipping, L1)	–	–	–	–	–	–	–	–
full model	57.95	–	0.731	36k	50.63	–	1.897	37k

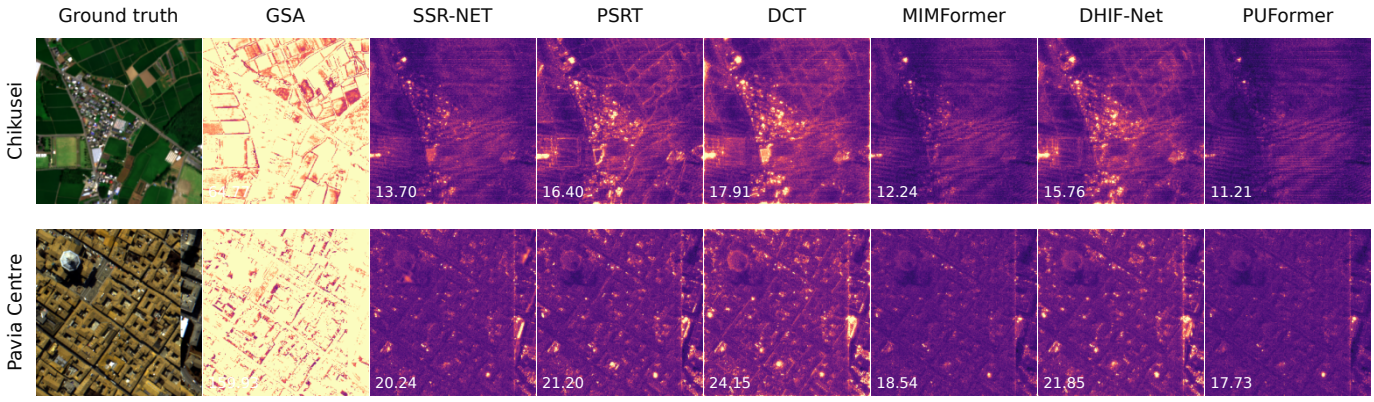


Fig. 7. Equal-budget benchmark: ground truth in false color and the mean absolute error over all bands (DN) of every method on the test image on which PUFormer has its lowest PSNR. The color scale is set by the retrained networks (clipped at the 99.5th percentile of their errors), so GSA saturates. The number in each map is its mean.

comes mainly from the eight ultraviolet bands (363–404 nm), which lie below the first WorldView-2 band and receive no high-resolution detail, and that the reported Q2n values do not follow PSNR the way our implementation does, so they were most likely computed differently. Because the RMSE is expressed in sensor digital numbers, it does not depend on any choice of peak, and it gives the most direct comparison across papers. Table S-IV of the supplementary material lists the long-trained models and the classical baselines under every PSNR and SSIM definition. Our GSA implementation scores 4.4 dB below the value reported in [15], so our protocol does not favor classical methods.

F. Comparison with reported results on Pavia Centre

Table X compares PUFormer with the 17 methods of [15] on Pavia Centre. Trained from scratch in one session, PUFormer reaches 50.69 dB PSNR, 3.53 dB above the best reported value, and lowers the RMSE by 34.1% and ERGAS by 25.3%. It ranks first on six of the eight metrics (PSNR, SSIM in the PSRT definition, ERGAS, CC, SCC and RMSE), but 13th on SAM (1.883° against 1.4542°) and 16th on Q2n (0.9753 against 0.9987). With the stricter per-image data range its SSIM is 0.9970, below the reported 0.9972. The PSNR gain is larger than on Chikusei although the training region is eight times smaller (97 against 800 non-overlapping patch equivalents).

The Q2n shortfall on this dataset, and part of the SAM shortfall, come from one test image that contains a stretch of the river Ticino (63% water pixels; Section S-IV of the supplementary material). Water reflects little light, so its spectra are dominated by sensor noise that neither observation carries: over water PUFormer's SAM is 2.95°, over all other pixels 1.56°, and this image alone carries 97% of the Q2n deficit of the four test images. Whether the test images of [15] contain water is not stated, so on this dataset SAM and Q2n are the least comparable of the eight metrics and RMSE the most. As on Chikusei, our GSA implementation scores below the value reported for it (34.78 against 35.35 dB), so the rebuilt

TABLE VI

WHERE THE ERROR LIES (THEOREM 2). OBS. MSE: MEAN SQUARED ERROR IN THE OBSERVED SUBSPACE $\mathrm{row}(\mathcal{D}) + \mathrm{row}(\mathcal{R})$, THE PART THAT CONTRADICTS THE OBSERVATIONS ($\times 10^{-6}$, NORMALIZED INTENSITIES). NULL: SHARE OF THE SQUARED ERROR IN THE FUSION NULL SPACE, WHICH ONLY A PRIOR CAN REDUCE. $\mathrm{PSNR}_{\mathcal{N}}$: THE PSNR THE METHOD WOULD REACH IF ITS OBSERVED PART WERE EXACT. EQUAL-BUDGET MODELS, SINGLE PASS.

	Chikusei $\times 4$				Pavia Centre $\times 4$			
Method	PSNR	Obs. MSE	Null (%)	$\mathrm{PSNR}_{\mathcal{N}}$	PSNR	Obs. MSE	Null (%)	$\mathrm{PSNR}_{\mathcal{N}}$
Bicubic	33.63	431.001	8.6	44.27	25.52	2745.835	11.5	34.91
GSA	43.20	46.899	13.7	51.92	34.78	369.778	18.7	42.39
SSR-NET	55.55	0.703	74.9	56.83	48.45	2.049	85.8	49.11
PSRT	53.99	1.645	64.0	56.01	47.87	2.483	85.4	48.56
DCT	53.18	2.240	58.2	55.60	46.94	4.312	79.5	47.94
MIMFormer	56.95	0.186	90.6	57.38	49.63	0.430	96.3	49.79
DHIF-Net	54.39	1.546	66.1	56.32	47.67	3.980	78.1	48.75
PUFormer	57.94	0.022	98.5	58.01	50.63	0.074	99.2	50.66

TABLE VII

EQUAL-BUDGET BENCHMARK ON THE OTHER DATASETS AND SCALE FACTORS: PSNR (DB) AND SAM (DEGREES) OF EVERY METHOD, RETRAINED UNDER THE SAME PROTOCOL AND 4 H BUDGET ON ONE T4 (SINGLE PASS). AT $\times s$ THE BLUR IS A $(2s-1)\times(2s-1)$ GAUSSIAN WITH $\sigma = s/2$; $\times 16$ TRAINS ON 128×128 PATCHES (BATCH 4). BEST LEARNED METHOD IN BOLD; –: RUN NOT AVAILABLE.

	CAVE $\times 4$		Harvard $\times 4$		Chikusei $\times 8$		Pavia Centre $\times 8$		Chikusei $\times 16$		Pavia Centre $\times 16$	
Method	PSNR↑	SAM↓	PSNR↑	SAM↓	PSNR↑	SAM↓	PSNR↑	SAM↓	PSNR↑	SAM↓	PSNR↑	SAM↓
GSA [4]	42.30	4.651	46.30	2.379	38.69	3.972	30.24	7.163	35.20	5.303	–	–
SSR-NET [8]	46.51	2.922	49.18	2.148	–	–	–	–	–	–	–	–
PSRT [10]	48.51	2.394	49.29	2.127	53.05	1.172	47.13	2.394	–	–	–	–
DCT [54]	46.36	2.794	48.92	2.188	51.77	1.318	45.51	2.739	–	–	–	–
MIMFormer [26]	50.93	1.955	49.72	2.061	55.57	0.919	49.09	2.064	54.21	1.029	–	–
DHIF-Net [20]	49.82	2.047	49.57	2.084	–	–	–	–	–	–	–	–
PUFormer	**52.57**	**1.653**	**50.01**	**2.012**	**56.82**	**0.822**	**49.94**	**1.986**	**55.89**	**0.885**	–	–

data do not favor our method.

G. Robustness of the long-trained models

The long-trained models of Tables IX and X behave like the ablation models under a blur mismatch (Table S-II and Fig. S3 of the supplementary material). With the training operator, a blur with σ between 1.5 and 3.0 costs 3.9–7.5 dB on Chikusei and 3.3–6.8 dB on Pavia Centre; with the true blur swapped in, it costs at most 0.26 and 0.11 dB. A spectral response shifted by 8 nm costs about 5 dB on Chikusei and is not compensated by swapping in the shifted response, because the priors rely on the training response beyond the data step; on Pavia Centre, whose IKONOS bands are 66–96 nm wide, the shift costs about 1 dB. Noise is the main weakness of models trained on noise-free data: at 30 dB SNR, PUFormer exceeds GSA by only 2.8 dB on Chikusei and by 6.0 dB on Pavia Centre.

H. One network for a family of observation models

Table XI tests the networks of Section IV-D on conditions that leave the training protocol: blur widths from $0.5\sigma_0$ to $2\sigma_0$, spectral responses shifted by up to 12 nm, noise from 40 to 30 dB SNR, all three at once and, outside the operator family, anisotropic and defocus blurs and an HR-MSI shifted against the LR-HSI by a fraction of a pixel (every condition is listed in the supplementary material). The operators are estimated from each test pair with Theorem 1. Within the family the estimate is exact: on every test image and condition, 30 dB SNR included, the blur width is recovered to within 0.719 and the spectral shift to within 0.1 nm on Chikusei (0.850 and 5.0 nm on Pavia Centre).

Three results stand out. First, training over the family is what makes the network robust to noise and to combined changes. At 40–30 dB SNR the mean PSNR rises from 49.08 to 52.91 dB on Chikusei and from 44.21 to 46.84 dB on Pavia Centre, and under combined changes from 48.12 to 52.83 dB and from 44.02 to 46.80 dB. The price is a lower PSNR at the training operator, 57.60 against 57.95 dB on Chikusei and 50.42 against 50.63 dB on Pavia Centre.

Second, within the family, giving the data steps the estimated operator adds to randomized training alone. With

TABLE VIII
OPERATOR ABLATION ACROSS DATASETS AND SCALE FACTORS (ONE RUN EACH, SEED 0, 4 H ON ONE T4; TEST PSNR IN DB). TRAINING BLUR: PSNR AT THE BLUR USED FOR TRAINING. CHANGED BLUR: WORST PSNR OVER THE THREE TEST BLURS $\sigma \in \{0.75, 1.25, 1.5\}\,\sigma_0$, WITH THE TRUE BLUR SUPPLIED (SWAP) OR THE TRAINING OPERATOR KEPT (FIXED). –: RUN NOT AVAILABLE.

	Training blur				Changed blur (worst case)				
Setting	Exact	Zero border	Learned	No data step	Exact, swap	Exact, fixed	Zero border, swap	Learned	No data step
Chikusei ×4	57.95	57.79	57.73	57.65	57.70	51.41	54.38	54.28	–
Pavia Centre ×4	50.63	50.52	50.57	50.46	50.50	43.13	49.64	49.33	–
CAVE ×4	52.57	52.19	52.22	51.69	52.29	48.66	51.82	51.08	51.38
Harvard ×4	50.01	49.97	49.98	49.84	49.95	48.41	49.29	49.69	49.74
Chikusei ×8	56.82	56.53	56.39	–	56.69	50.00	55.27	55.31	–
Pavia Centre ×8	49.94	49.64	49.56	–	49.88	42.69	49.38	48.43	–

TABLE IX
CHIKUSEI ×4 COMPARISON. ALL ROWS EXCEPT PUFORMER ARE REPORTED IN [15]. BEST VALUE PER COLUMN IN BOLD. SSIM OF PUFORMER USES THE PSRT DEFINITION (DATA RANGE 1); WITH DATA RANGE EQUAL TO EACH IMAGE'S MAXIMUM IT IS 0.9969 (SINGLE PASS) AND 0.9970 (SELF-ENSEMBLE).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [4]	47.5769	0.9933	1.7216	2.1667	0.9658	0.9894	0.9773	70.54
CNMF [5]	44.5291	0.9891	1.7259	2.8676	0.9662	0.9828	0.9645	101.64
FUSE [7]	41.1550	0.9787	2.5814	3.4710	0.9418	0.9736	0.9486	148.10
U2Net [34]	50.5455	0.9956	1.2389	1.8740	0.9928	0.9920	0.9798	49.97
PSRT [10]	50.6722	0.9952	1.2733	2.0107	0.9862	0.9906	0.9778	49.00
DCT [54]	52.1998	0.9977	0.9239	1.7448	0.9832	0.9935	0.9873	41.45
DDIF [14]	53.4421	0.9972	0.9387	1.8087	0.9963	0.9929	0.9840	35.58
MIMFormer [26]	53.3578	0.9973	0.9094	1.7611	0.9948	0.9931	0.9850	35.90
MIMO-SST [11]	53.6073	0.9972	0.9315	1.8119	0.9944	0.9926	0.9834	34.93
OTPNet [35]	52.8243	0.9969	1.0201	1.9332	0.9940	0.9916	0.9841	38.49
SMGU [39]	54.5244	0.9977	0.8719	1.6405	0.9952	0.9939	0.9875	31.29
AMSF [27]	49.3827	0.9944	1.4852	2.3943	0.9878	0.9879	0.9790	56.78
SSDT [13]	52.4788	0.9975	0.8768	1.7992	0.9939	0.9928	0.9847	40.00
ASMNet [28]	53.4528	0.9972	0.9472	1.9153	0.9874	0.9917	0.9848	35.89
DPFormer [29]	53.1632	0.9970	0.9483	1.8795	0.9949	0.9923	0.9824	37.18
CLSNet [53]	55.2007	0.9979	0.7836	1.6318	0.9943	0.9941	0.9869	29.06
Two-stage diffusion [15]	56.1871	0.9982	0.7203	1.5158	**0.9969**	0.9948	0.9891	25.73
PUFormer (single pass)	58.1467	**0.9989**	0.7171	1.3367	0.9930	**0.9967**	**0.9992**	19.06
PUFormer (self-ensemble)	**58.2016**	**0.9989**	**0.7126**	**1.3304**	0.9931	**0.9967**	**0.9992**	**18.94**

observation-model conditioning the network is the most accurate in every group, for instance 52.83 against 52.30 dB for the same network trained on randomized data with the nominal operator under combined changes on Chikusei, and 46.80 against 46.35 dB on Pavia Centre. Observation-model conditioning adds little to operator conditioning alone (52.91 against 52.79 dB under noise on Chikusei), and the exact operator stays ahead of the zero border (52.70 against 52.45 dB).

Third, outside the family the conditioned network is fragile. With an anisotropic or a defocus blur, the estimated isotropic operator is wrong; the conditioned network, which trusts its operator, falls to 54.78 dB on Chikusei and 47.02 dB on Pavia Centre, whereas the network trained on randomized data with the nominal operator keeps 57.18 and 50.01 dB; given the true kernel, the conditioned network recovers (57.31 and 50.29 dB). A sensitivity test shows why. With noise-free nominal data and a blur of $0.9\,\sigma_0$ in its data steps, a relative operator residual $\|(\tilde{\mathcal{D}} - \mathcal{D})\mathbf{X}\|/\|\mathcal{D}\mathbf{X}\|$ of only 0.7%, the conditioned network drops to 57.01 dB on Chikusei and the nominally trained one to 56.30 dB, while the randomized-only network is unaffected (57.42 dB): as Theorem 3 states, an exact data step trusts its operator. Exactness thus makes the network precise when the operator family contains the test conditions and fragile when it does not, so the family used for training and estimation has to cover the sensors it will meet. Finally, an HR-MSI misregistered by half a pixel costs every method more than 15 dB (38.15 dB on Chikusei, with GSA at a similar loss): none of the methods compared here models registration errors.

I. Complexity

Table XII compares the parameters, operations, inference time and memory of all retrained networks on the same input. PUFormer has 10.13 M parameters and needs 296 GFLOPs for a 256×256 Chikusei image, which it reconstructs in 0.18 s on one Tesla T4 with mixed precision (92 bands: 0.18 s); the self-ensemble doubles this. It is slower than SSR-NET, MIMFormer and PSRT, but more than ten times faster than DCT and DHIF-Net. Because the attention is computed across channels, memory and time grow linearly with the number of pixels.

VII. DISCUSSION AND LIMITATIONS

What the exact operators buy. The exact operators matter less for accuracy at the training operator than for what they

TABLE X

PAVIA CENTRE ×4 COMPARISON. ALL ROWS EXCEPT PUFORMER ARE REPORTED IN [15] ON THE BENCHMARK DATA RELEASED WITH PSRT [10]; PUFORMER IS EVALUATED ON OUR SIMULATION OF THE SAME PROTOCOL FROM THE ORIGINAL SCENE (SECTION V). BEST VALUE PER COLUMN IN BOLD. SSIM OF PUFORMER USES THE PSRT DEFINITION (DATA RANGE 1); WITH DATA RANGE EQUAL TO EACH IMAGE'S MAXIMUM IT IS 0.9970 (SINGLE PASS) AND 0.9970 (SELF-ENSEMBLE).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [4]	35.3495	0.9736	5.0812	2.7260	0.9818	0.9832	0.9768	137.78
CNMF [5]	32.1693	0.9469	3.8494	3.8446	0.9489	0.9683	0.9294	198.26
FUSE [7]	34.7418	0.9612	5.5628	3.2042	0.9103	0.9779	0.9469	146.90
U2Net [34]	44.3439	0.9963	1.6542	1.0924	0.9964	0.9973	0.9963	49.10
PSRT [10]	45.8875	0.9970	1.5230	0.9750	0.9974	0.9978	0.9966	41.85
DCT [54]	46.6745	0.9971	1.5169	0.8753	0.9973	0.9983	0.9972	37.66
DDIF [14]	46.2530	0.9970	1.5253	0.9125	0.9960	0.9982	0.9970	39.56
MIMFormer [26]	44.8497	0.9959	1.8225	1.0683	0.9957	0.9975	0.9954	46.19
MIMO-SST [11]	46.6334	0.9969	1.5587	0.8607	0.9975	0.9984	0.9972	37.68
OTPNet [35]	45.8081	0.9967	1.6573	0.9465	**0.9987**	0.9980	0.9968	41.51
SMGU [39]	47.1239	0.9972	1.4690	0.8547	0.9981	0.9984	0.9974	37.09
AMSF [27]	43.4465	0.9950	2.0533	1.2728	0.9929	0.9965	0.9940	54.25
SSDT [13]	46.3173	0.9965	1.6439	0.8918	0.9968	0.9982	0.9969	39.01
ASMNet [28]	46.2072	0.9966	1.6517	0.9019	0.9940	0.9982	0.9969	39.50
DPFormer [29]	42.4280	0.9948	1.9367	1.3986	0.9945	0.9957	0.9917	60.96
CLSNet [53]	46.5277	0.9969	1.5708	0.8412	0.9983	0.9983	0.9975	36.21
Two-stage diffusion [15]	47.1597	0.9972	**1.4542**	0.8262	0.9980	0.9985	0.9976	35.71
PUFormer (single pass)	50.6299	**0.9974**	1.8955	0.6206	0.9751	**0.9994**	**0.9995**	23.68
PUFormer (self-ensemble)	**50.6915**	**0.9974**	1.8833	**0.6170**	0.9753	**0.9994**	**0.9995**	**23.52**

TABLE XI

ONE NETWORK FOR A FAMILY OF SENSORS: MEAN TEST PSNR (DB) OVER GROUPS OF TEST CONDITIONS THAT DIFFER FROM THE TRAINING PROTOCOL. RANDOMIZED: TRAINED WITH A BLUR, SRF SHIFT AND NOISE LEVEL DRAWN ANEW FOR EVERY BATCH; + OPERATOR: THE BATCH'S OPERATOR IS ALSO GIVEN TO THE DATA STEPS. FIXED: THE TRAINING-PROTOCOL OPERATOR IS KEPT AT TEST TIME; EST.: BLUR AND SRF SHIFT ESTIMATED FROM THE TEST PAIR ALONE; TRUE: THE TRUE TEST OPERATOR (AN UPPER REFERENCE, NOT MARKED). †ANISOTROPIC AND DEFOCUS BLURS, WHICH THE OPERATOR FAMILY DOES NOT CONTAIN; ‡AN HR-MSI SHIFTED BY 0.5–1 PIXEL AGAINST THE LR-HSI. BEST PER ROW IN BOLD.

	MIMFormer, nominal training	MIMFormer, randomized	PUFormer, nominal training		PUFormer, randomized data only		PUFormer, randomized, zero border	PUFormer, randomized + operator			+ observation-model conditioning		
Test condition	fixed	fixed	fixed	est.	fixed	est.	est.	fixed	est.	true	fixed	est.	true
Chikusei ×4													
Training operator	56.96	56.50	**57.95**	**57.95**	57.43	57.43	57.46	57.59	57.59	57.59	57.60	57.60	57.60
Blur ×0.5–2	54.18	56.38	50.58	57.02	57.33	57.13	57.36	53.69	57.49	57.49	53.68	**57.51**	57.51
SRF shift ±4–12 nm	53.61	55.66	50.02	54.29	57.35	55.13	57.38	51.45	57.51	57.51	49.71	**57.54**	57.54
Noise 40–30 dB	50.33	51.67	49.08	49.08	52.35	52.35	52.52	52.79	52.79	52.79	**52.91**	**52.91**	52.91
Blur + SRF + noise	48.98	51.30	45.10	48.12	52.30	51.12	52.45	48.34	52.70	52.70	46.53	**52.83**	52.83
Worst of the above	45.78	48.03	42.30	44.35	49.25	48.12	49.49	45.99	49.80	49.80	44.09	**49.98**	49.98
Blur outside the family†	54.23	56.43	50.16	52.54	**57.18**	57.07	56.39	53.59	54.79	57.41	53.57	54.78	57.31
LR-HSI grid offset	–	–	–	–	–	–	–	–	–	–	–	–	–
HR-MSI misregistered‡	38.00	37.46	**38.38**	38.34	37.48	37.48	37.82	38.15	38.12	–	38.15	38.12	–
Pavia Centre ×4													
Training operator	49.63	49.37	**50.63**	**50.63**	50.40	50.40	50.41	50.45	50.45	50.45	50.42	50.42	50.42
Blur ×0.5–2	47.05	49.28	42.33	49.77	50.33	50.22	50.36	45.85	**50.40**	50.40	46.19	50.35	50.35
SRF shift ±4–12 nm	48.66	48.31	49.68	49.72	50.32	49.39	50.35	49.59	**50.39**	50.39	49.00	50.37	50.37
Noise 40–30 dB	45.24	45.87	44.21	44.21	46.40	46.40	46.51	46.76	46.76	46.76	**46.84**	**46.84**	46.84
Blur + SRF + noise	44.15	45.22	40.85	44.02	46.35	45.90	46.49	44.57	46.72	46.72	44.64	**46.80**	46.80
Worst of the above	41.35	42.63	35.56	40.40	43.69	43.29	43.91	40.48	44.27	44.27	41.38	**44.39**	44.39
Blur outside the family†	46.89	49.32	41.48	44.00	**50.01**	49.89	48.65	45.45	46.77	50.41	45.87	47.02	50.29
LR-HSI grid offset	–	–	–	–	–	–	–	–	–	–	–	–	–
HR-MSI misregistered‡	29.86	29.33	**30.39**	30.35	29.50	29.52	29.75	30.09	30.07	–	30.03	30.01	–

TABLE XII

COMPLEXITY FOR ONE 256 × 256 CHIKUSEI IMAGE (128 BANDS, ×4) ON ONE TESLA T4.

Method	Params (M)	GFLOPs	Time (ms)	Memory (GB)
PUFormer	10.13	296	178	0.44
SSR-NET	0.44	58	11	0.11
PSRT	0.30	43	118	0.42
DCT	8.46	2640	2103	1.10
MIMFormer	16.46	224	62	0.45
DHIF-Net	48.67	14930	2938	2.08

make possible. Their separable structure gives the data step in closed form and the null space of fusion explicitly, which turns error analysis into a split between what contradicts the observations and what only a prior can supply; PUFormer's error in the observed subspace is several times smaller than that of every retrained network. Because the operator enters only the data steps, it can be supplied, estimated from the test pair and replaced, and a single network covers a family of blurs, responses and noise levels. The same property makes the network depend on its operator: inside the family this is a strength, outside it a weakness, and the family used for training and estimation has to cover the sensors met in practice. The prior, in contrast, is a building block rather than a contribution: most of the margin over the retrained networks is already reached by the residual U-Net alone, and a convolutional mixer performs as well as transposed attention.

Limitations. All quantitative results with a reference rest on Wald's protocol, in which the observations are simulated from the reference, as in every benchmark of this field; in real data the border rule, the blur and the spectral response are only approximately known, and Theorems 1 and 2 assume that the operator family contains the true operator and that the blur is separable. Section VI-H shows the consequence: outside the family, an operator-conditioned network that trusts a wrong operator loses several decibels, more than a network trained

on randomized data without an operator, and a misregistration of half a pixel defeats every method compared here. Wider operator families, learned with the network or covered by the training distribution, and an explicit registration step are needed for real sensor pairs. The equal-budget benchmark gives every method 4 h on one GPU; the authors' own recipes do not improve the baselines under this budget, and an 8 h check narrows but keeps the gap, but fully converged comparisons were not run. The 17-method comparison of Tables IX and X uses copied values on test images that follow the published protocol but are not the same pixels; our retrained baselines exceed their reported values by up to 3.6 dB, so these tables give context rather than a ranking, and the strongest reported methods could not be retrained because their code is not available. The test sets of Chikusei and Pavia Centre are small (8 and 4 images), so significance is assessed on the pooled settings.

VIII. Conclusion

We formulated hyperspectral and multispectral image fusion as maximum a posteriori estimation with parametric operators and unrolled it, and we proved three properties of the operators inside such a network: their parameters are identified by the observations, their separable structure gives the data step in closed form together with the null space of fusion, and an exact data step leaves the ground truth fixed while an inexact one moves it by a bounded amount. PUFormer, built on these results, is the most accurate of six networks retrained under one budget on Chikusei and Pavia Centre, its error lies almost entirely in the null space that no observation constrains, and trained over a family of observation models with estimated operators it keeps its accuracy under changed blurs, spectral responses and noise. Operator families that cover non-separable blurs and registration errors, and real sensor pairs, are the main directions for future work.

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